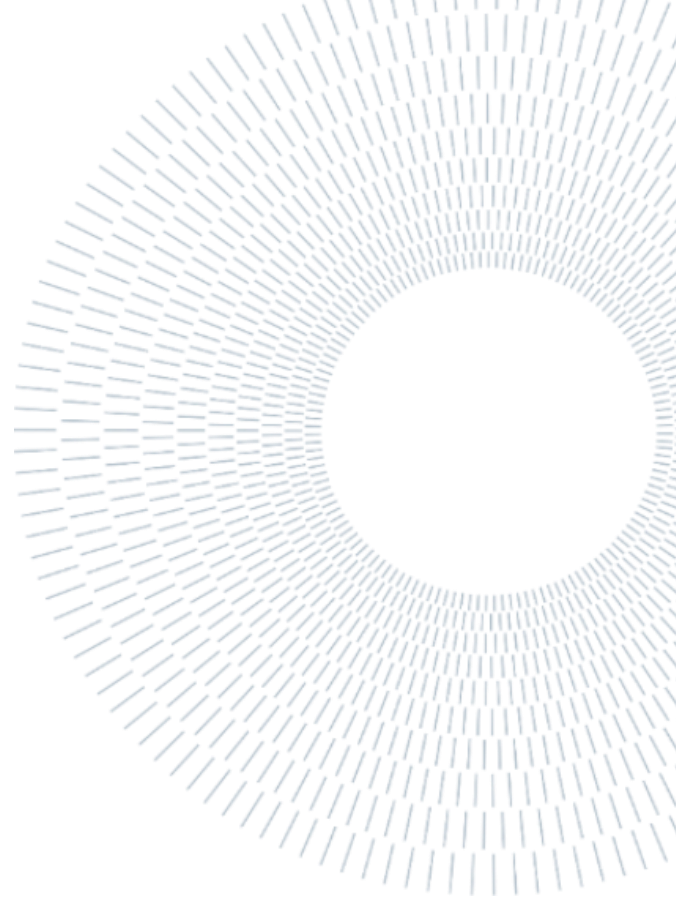




POLITECNICO
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SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Master of Science in Mechanical Engineering -
Sports Engineering (CC6)



Analysis of motorcycle dynamics and stability

“Dynamics of mechanical systems” – Project Review

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1. Introduction

This project aims to analyze the complex behavior of motorcycles under excitations and test different operating conditions, with the objective of identifying the configurations that best guarantee rider safety, vehicle control and stability.

During time, the manufacturers developed very sophisticated devices to damp oscillations (like weaving and wobbling), prevent sliding provoked by lateral forces by increasing tires grip or progressive suspensions to prevent geometry from changing too abruptly during braking/acceleration.

This study focuses on a real experimental acquisition performed with the intention to excite a motorcycle on different racing speeds, allowing to collect data for the analysis through mathematical tools such as Gauss Diagram and FRF computation.

The main event of interest is the Weave vibration, a low-frequency (2–4 Hz) global eigenmode characterized by coupled oscillations in roll, yaw and steer involving the entire vehicle chassis. Typically emerging at medium-to-high speeds, it is driven by the interaction between frame stiffness, tire properties and aerodynamics, often exhibiting reduced damping as velocity increases.

2. State of the art

The general review of the literature in the context of dynamics behavior and stability of motorcycles, indicates three main modes of vibration (eigenvalues):

1. *Weave*
2. *Wobble*: high-frequency oscillation (6–10 Hz) of the front assembly around the steering axis.
3. *Capsize*: non-oscillatory eigenmode dominated by the motorcycle's roll motion. It describes the vehicle's natural tendency to fall sideways away from the vertical equilibrium.

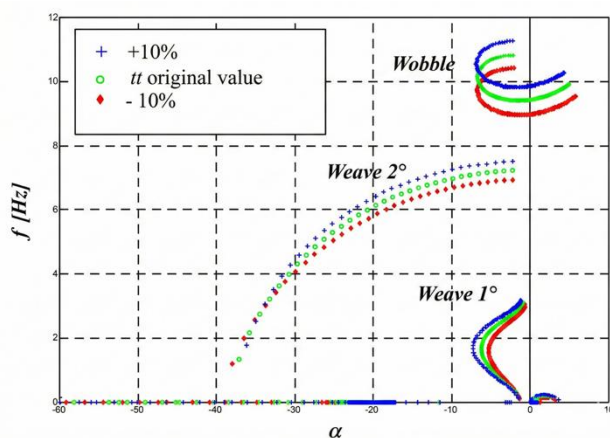


Figure 1: Gauss Diagram

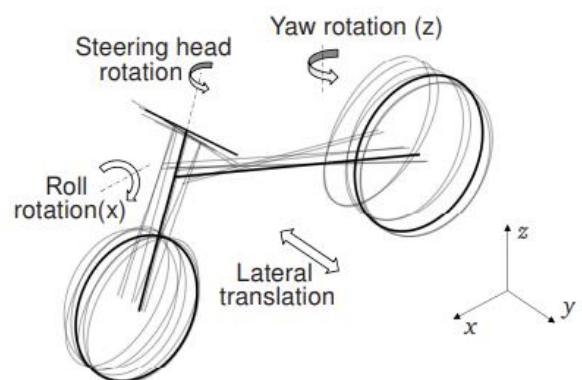


Figure 2: Weave motions [2]

These are important parameters to keep under control with the aim of guaranteeing safety on the streets to protect drivers from unexpected impacts or imbalances that could cause incidents.

In the context of this project, the analysis focuses primarily on isolating and evaluating the weave mode. As defined in the literature, the weave mode is a low-frequency oscillation involving the coupled yaw and roll motion of the entire vehicle. In his review [1] R.S. Sharp refined the understanding of this mode, showing that it is generally well-damped at medium speeds but can become unstable at low velocities or transition into oscillatory instability at very high speeds. The isolation of this mode is critical because it represents the primary handling limit for the vehicle's trajectory control.

Conversely, the wobble mode is intentionally deprioritized in this study. The justification for this exclusion is twofold. Firstly, as demonstrated by R.S. Sharp, an accurate representation of the wobble mode requires the inclusion of tire relaxation properties, as the delay in lateral force generation is a governing factor in high-frequency steering oscillations.

Secondly, recent studies by F. Passigato [2] indicate that the wobble mode is strongly influenced by structural flexibilities, which require complex elastic multibody simulations (EMBS) rather than rigid-body models. Furthermore, Dębowski [3] highlights that wobble sensitivity is heavily dependent on specific mass parameters and steering inertia, which are secondary to the global stability parameters investigated here.

Following the methodology standardized by R.S. Sharp and expanded in modern studies, the results are visualized using a Gauss diagram (Root Locus plot). This approach allows for the immediate identification of stability boundaries, where the migration of eigenvalues into the positive real half-plane indicates the onset of instability. By mapping the trajectory of the weave mode roots, this methodology provides a robust metric for assessing the vehicle's dynamic safety.

The dynamic stability of vehicles represents a complex engineering challenge due to the inherent coupling between the operational modes of the rider, the chassis, and the tire-road interface. The foundation for modern stability analysis was established by R.S. Sharp [4] in 1971, whose work demonstrated that the lateral dynamics of a motorcycle can be adequately described by a mathematical model considering four primary degrees of freedom (4-DOF): lateral displacement (γ), yaw angle (ψ), roll (ϕ), and the forecarriage rotation around the steer axis (δ). Sharp's analysis highlighted that while higher-order models are necessary for investigating high-frequency phenomena, the 4-DOF model is sufficient to capture the dominant eigenmodes governing the vehicle's global stability, specifically the "capsize" and "weave" modes.

3. Experimental analysis

A MATLAB® code has been written to perform all the analysis and obtain graphical representations of the most relevant results.

3.1 Acquisition overview

The sensors used for the detection of motorcycle behavior over the path were applied as shown in the figure.

The main sensors applied were:

- Gyroscope on handlebar (voltmeter): to measure angular speed on steering axis.
- Gyroscope on the rear: to measure angular speed on yaw axis.
- GPS: to obtain latitude and longitude.

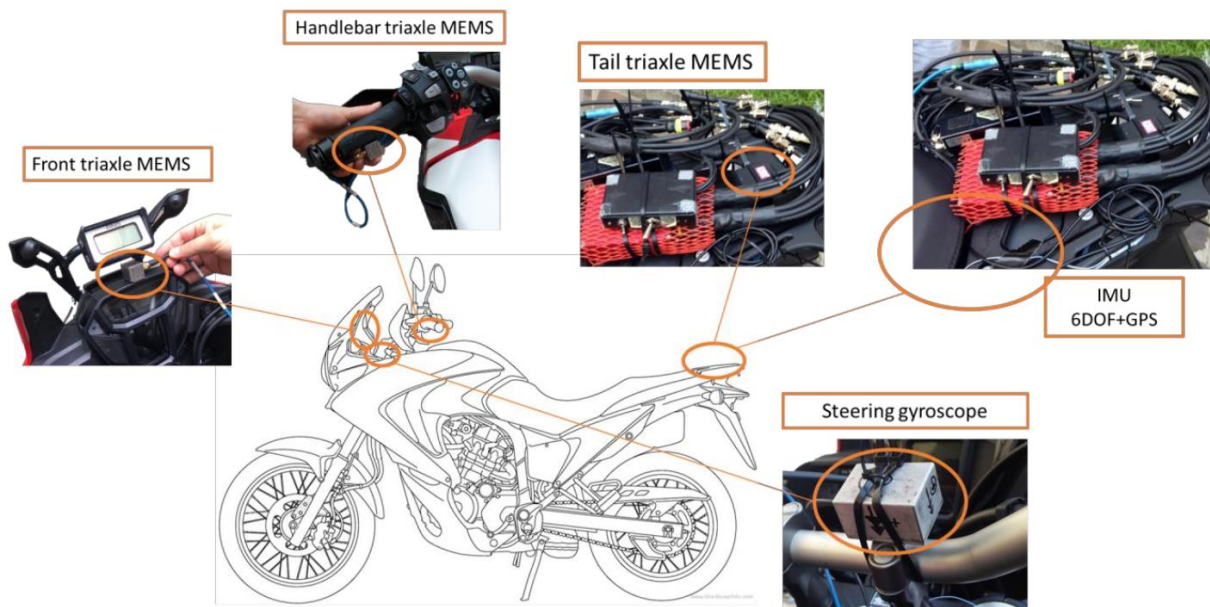


Figure 3: Sensors disposition

The following graphs show the signals achieved from the MEMS the bike is equipped with.

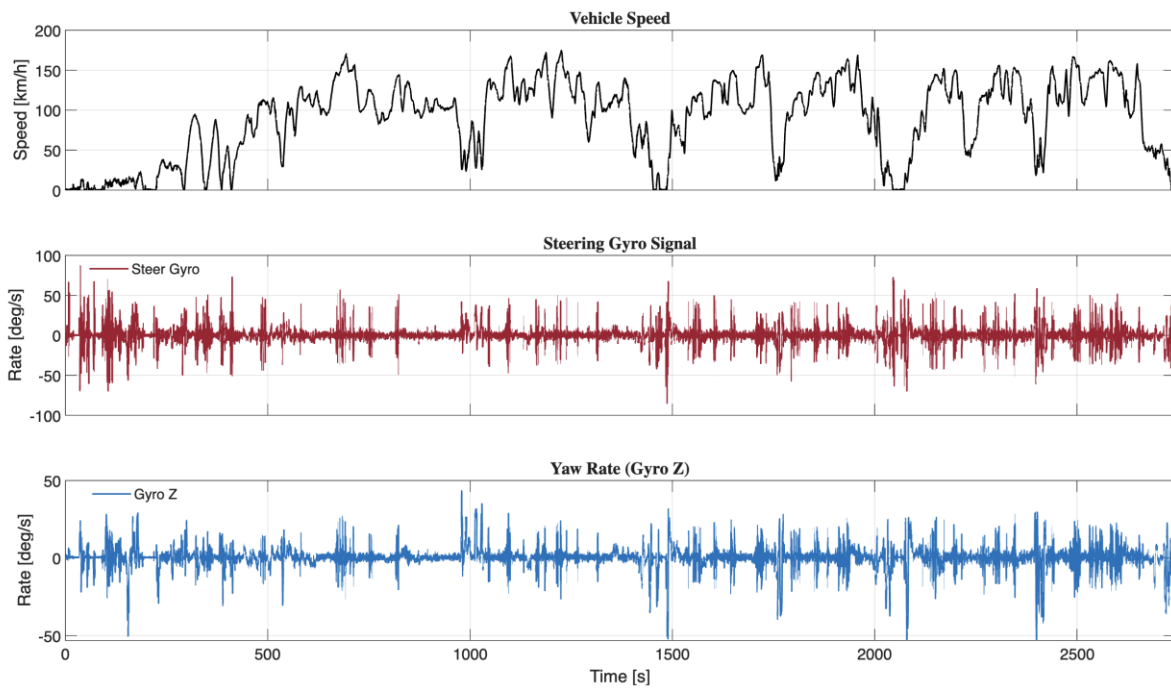


Figure 4: Global signals: Vehicle Speed, Steering Gyro Signal (obtained by the signal in Volt, using the Sensitivity provided), Yaw Rate

In the “Vehicle Speed” plot is evident that there are two moments when the vehicle stops, this is a noticeable element for subsequent considerations.

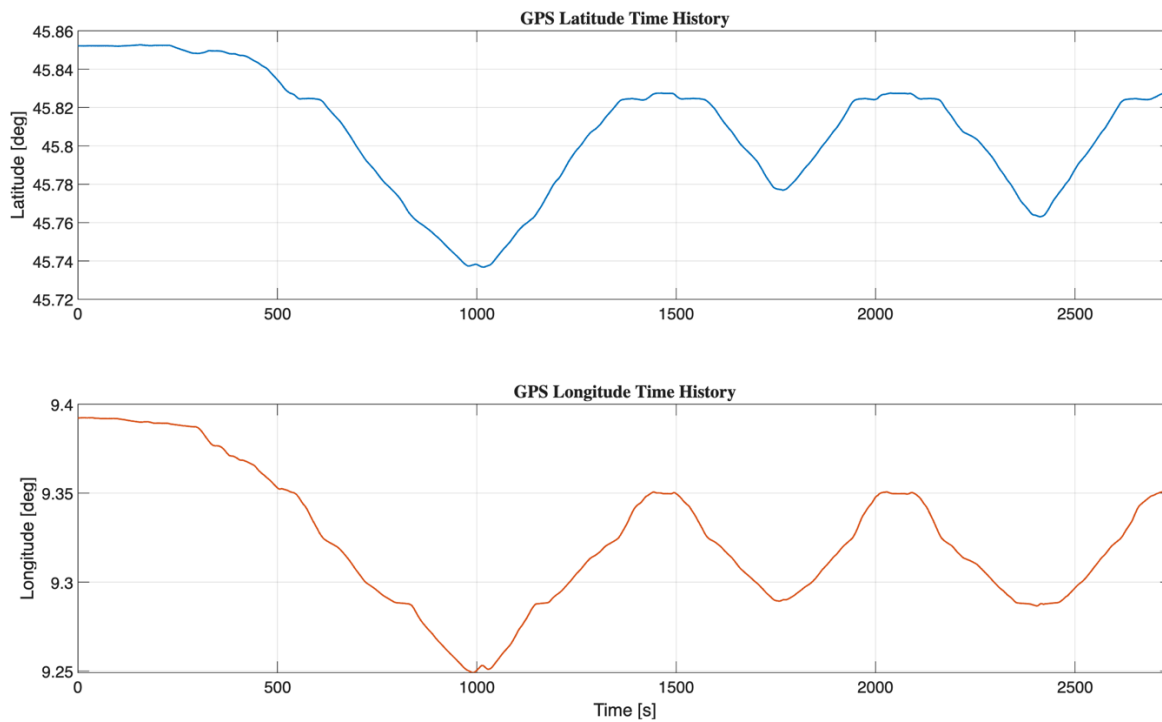


Figure 5: GPS Longitude and Latitude

The path has been obtained by merging peaks in the signals of latitude and longitude, that indicated the presence of 6 different segments executed during the acquisition.

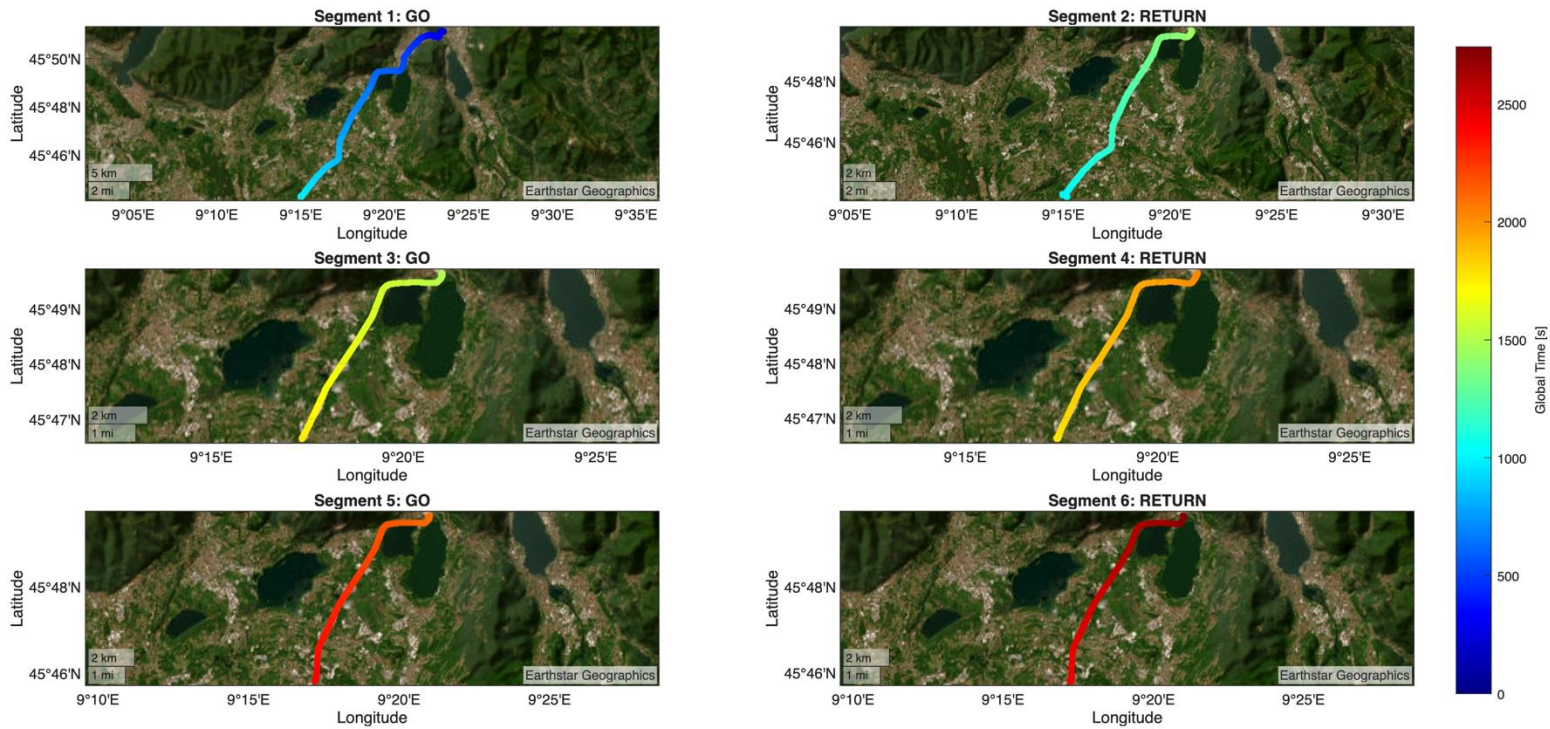


Figure 7: Segments Geoplot

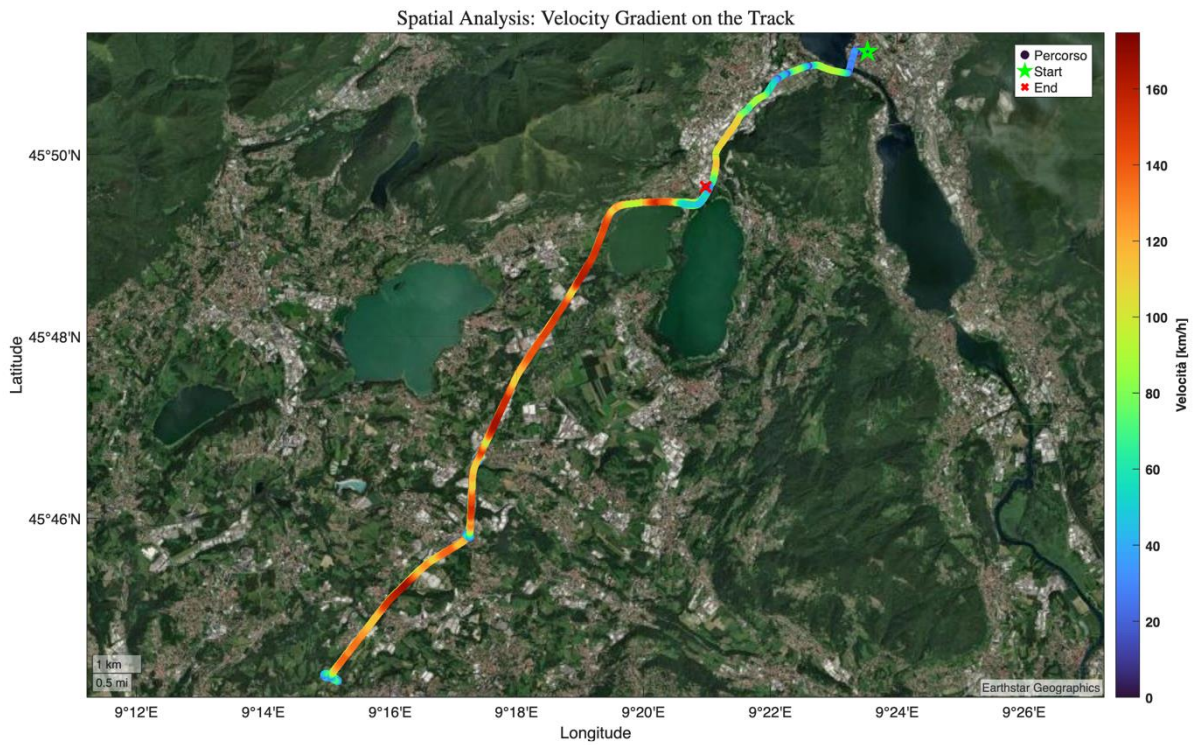


Figure 6: Speed gradient Geoplot

3.2 Data analysis

3.2.1 Impulse detection and processing

Throughout the test run, multiple excitation events were introduced by the rider via controlled body motions, resulting in transient dynamic responses of the motorcycle.

The detection of these impulses has been performed by manual selection from the user through 6 different interactive windows that show two graphs simultaneously: the overlap of Steer signal and Gyro_z signal compared with the speed during the section.

The gyroscope signals have been previously filtered using a high-pass filter with a cutoff frequency of 2 Hz in order to remove low-frequency noise and drift.

Before selection:

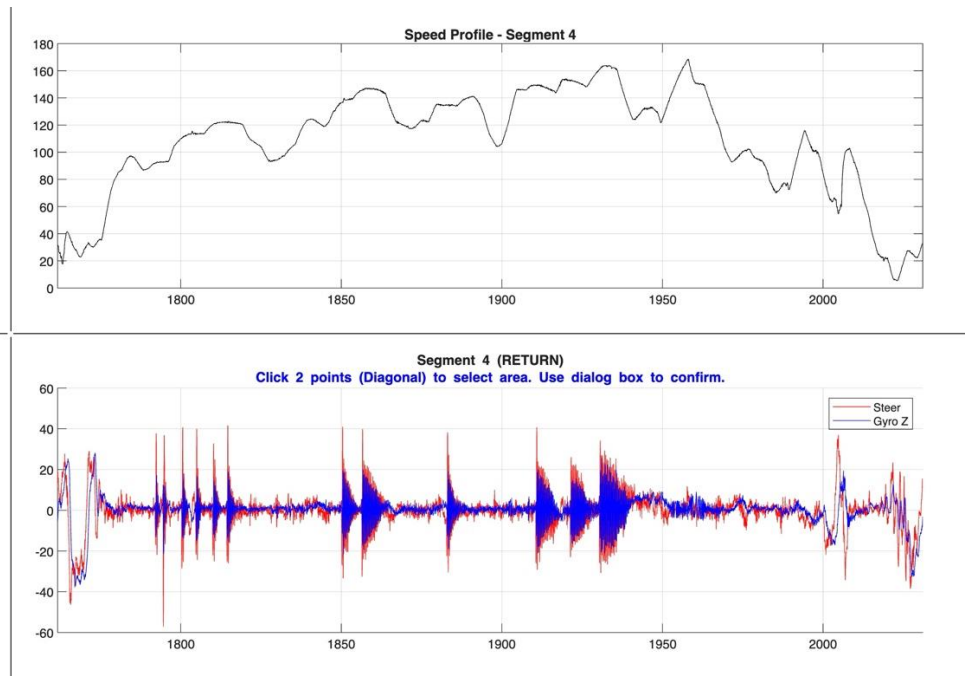
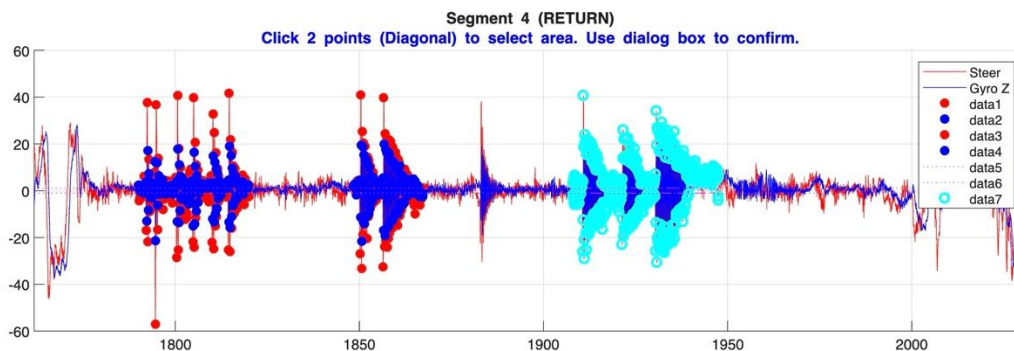


Figure 8: Example of peaks selection

After selection:



Observing the 6 sections it is evident that the driver increased the speed by steps in the rectilinear zone of the path and imposed the excitations at different values of velocity.

All the peaks are collected in vectors associated with the speed and the instant of time they occur and then they are approximated by the Hilbert Transform, as shown in the figure below.

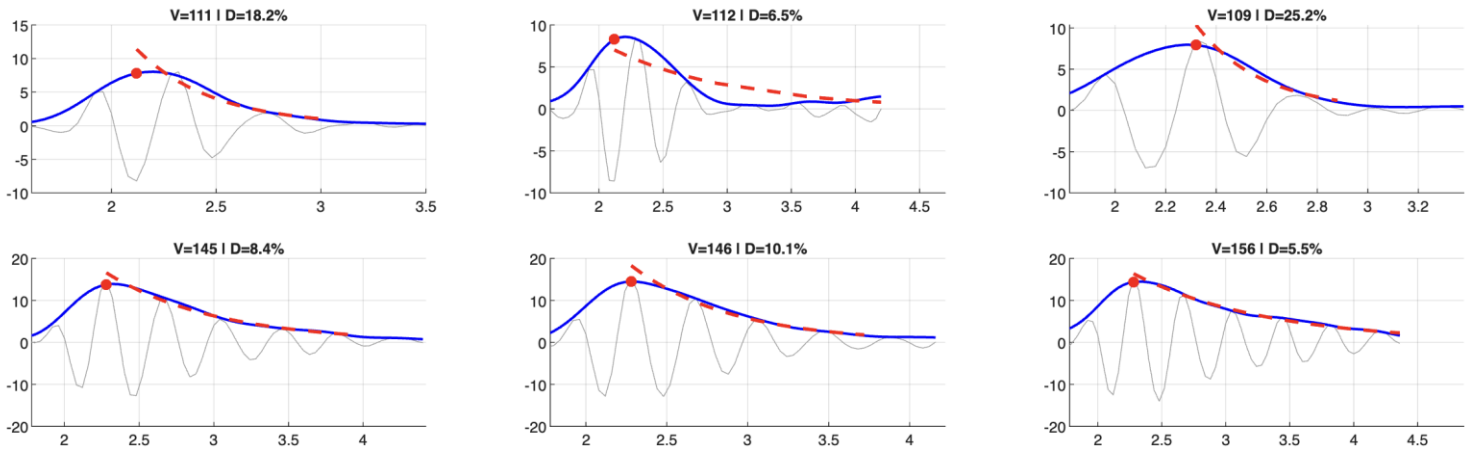


Figure 9: Examples of Hilbert Transform reconstruction

The aim of the reconstruction is to simplify the evaluation of the damping trend for each peak, that consists in the calculation of the modal damping ratio and the decay rate for each of those.

Before the actual numerical evaluation of the damping parameters, it is necessary to perform the Fast Fourier Transform for every peak detected to define the damped natural frequency (ω_m), necessary to calculate other parameters.

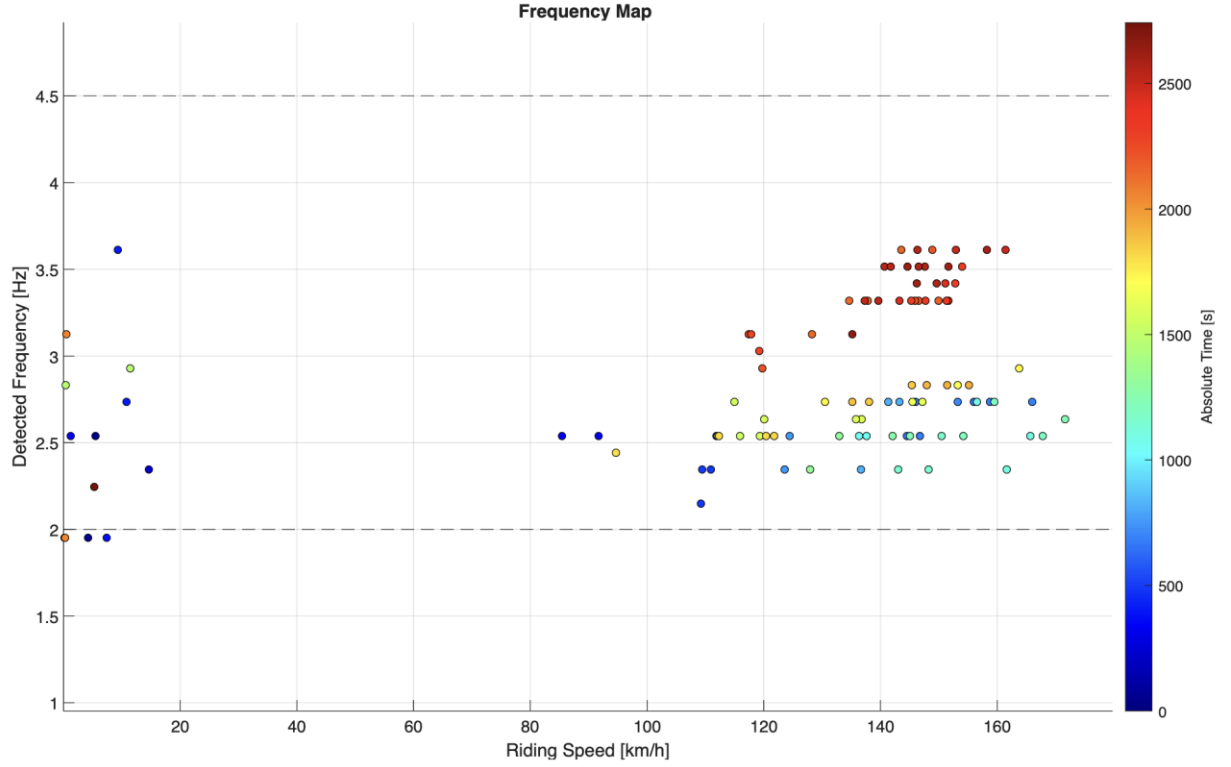


Figure 10: Frequency vs Riding speed

The figure below shows the results obtained:

This plot reveals a distinct temporal trend within the weave bandwidth: lower frequencies are predominantly associated with the initial phase of the experiment, while a progressive shift toward higher frequencies is observed as the session proceeds.

The decay rate (σ) is extracted by performing a linear regression on the natural logarithm of the instantaneous amplitude envelope, effectively linearizing the exponential decay of the oscillation. This parameter quantifies the temporal rate of energy dissipation and is essential for calculating the modal damping ratio (ζ), the definitive metric for evaluating the vehicle's dynamic stability.

The modal damping ratio (ζ) of the weave mode is estimated through a multi-stage signal processing approach designed to reduce noise sensitivity and non-stationarity effects. The logarithm of the envelope is linearly regressed to estimate the decay rate (σ), using the full decay window rather than discrete peaks.

The damping ratio is finally computed as:

$$V = \frac{s}{\sqrt{s^2 + w_d^2}} * 100$$

ω_d = dominant damped natural frequency.

The results have been collected in the following plot that reveal an exponential decay of both the metrics:

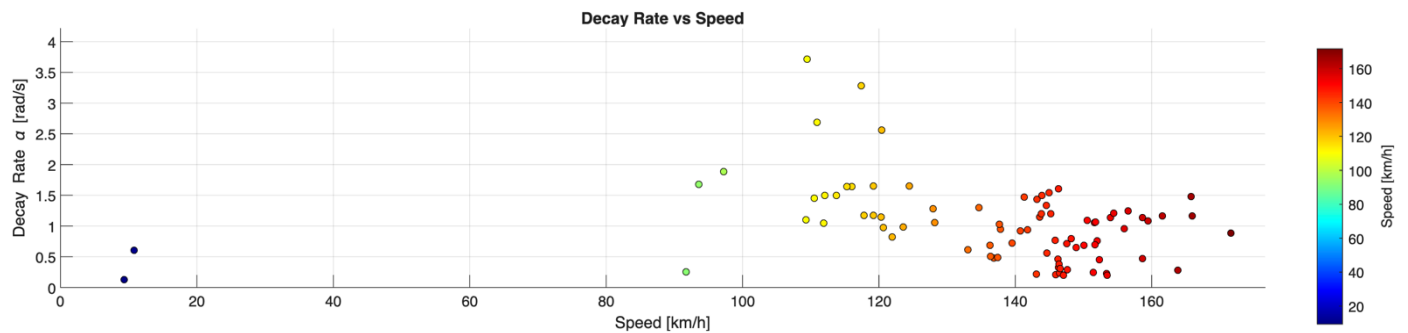


Figure 11: Decay Rate (Rad/s)

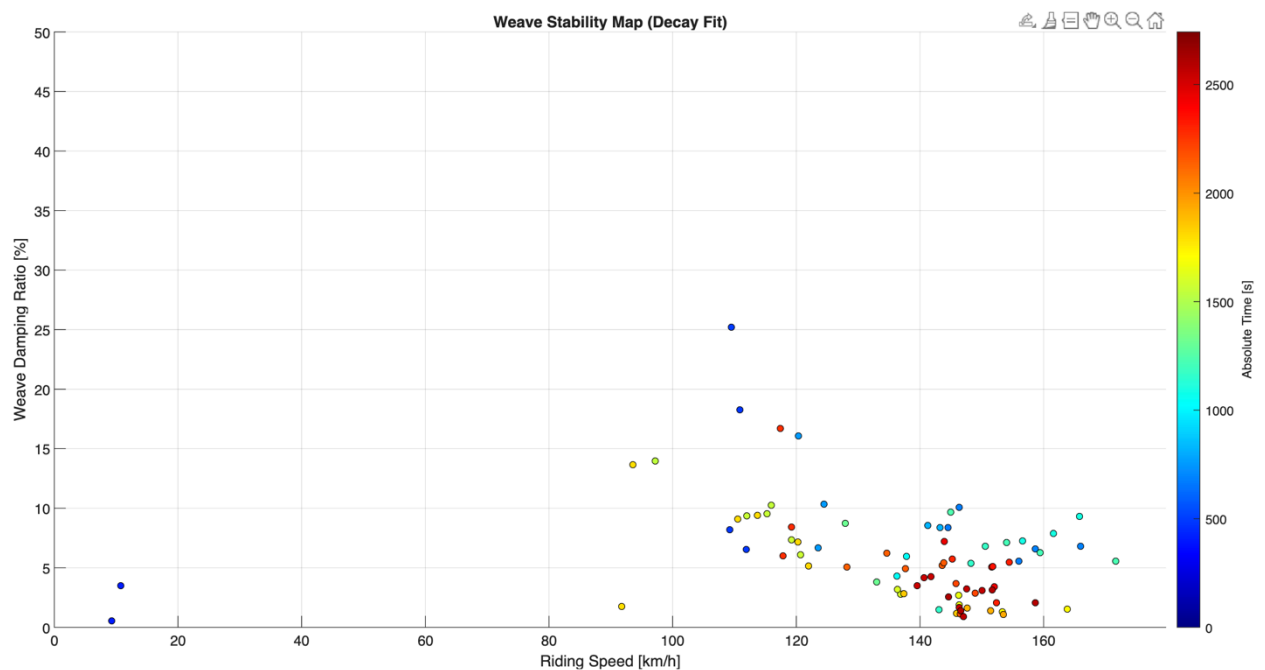


Figure 12: Damping Ratio (%)

In Figure 11 and 12 is important to notice that the data points corresponding to the final phase of the experiment lie systematically lower than those associated with the initial acquisition, even at comparable vehicle speeds. This trend indicates a progressive reduction in the motorcycle's ability to dissipate dynamic excitations as the test proceeds. This is coherent with the previous observations about the shift in frequencies

revealed by the plot in figure 10. The possible explanations of this phenomenon are reported in the next paragraphs.

To confirm these relations, a further analysis has been performed to show the distribution of frequencies during the 6 sections, that confirms the increase of frequencies in the last sectors.

Weave Identification by Sector (Auto-Spectrum)

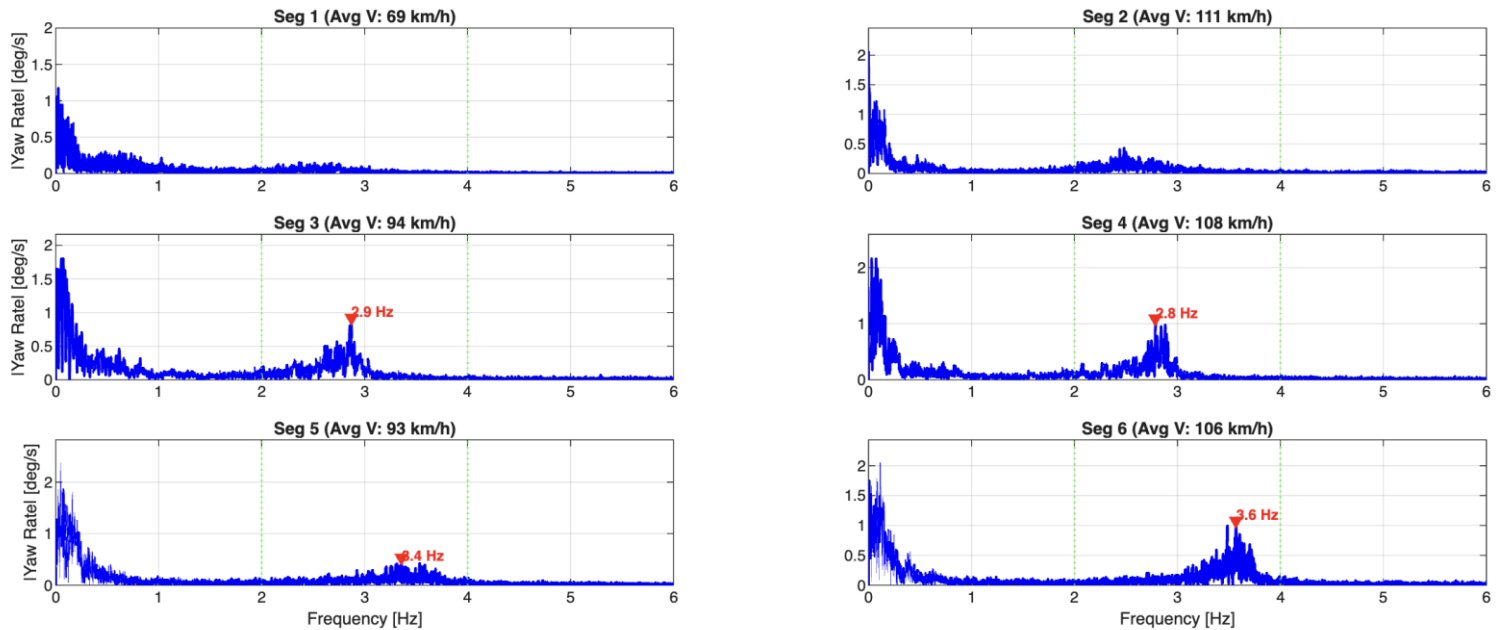


Figure 13: Spectral content divided by section

3.2.2 Statistical analysis

Given the stochastic nature of experimental road testing, caused by variability in rider inputs, environmental conditions, and unavoidable sensor noise, a statistical approach was adopted to characterize the vehicle's dynamics in a more trustable way.

Individual damping and frequency estimates were clustered into speed intervals of 10 km/h (starting from the first signal detected). For each interval, the central tendency (mean) was computed to identify the stability trend versus speed, while the standard deviation was analyzed to quantify the dispersion and repeatability of the experimental data.

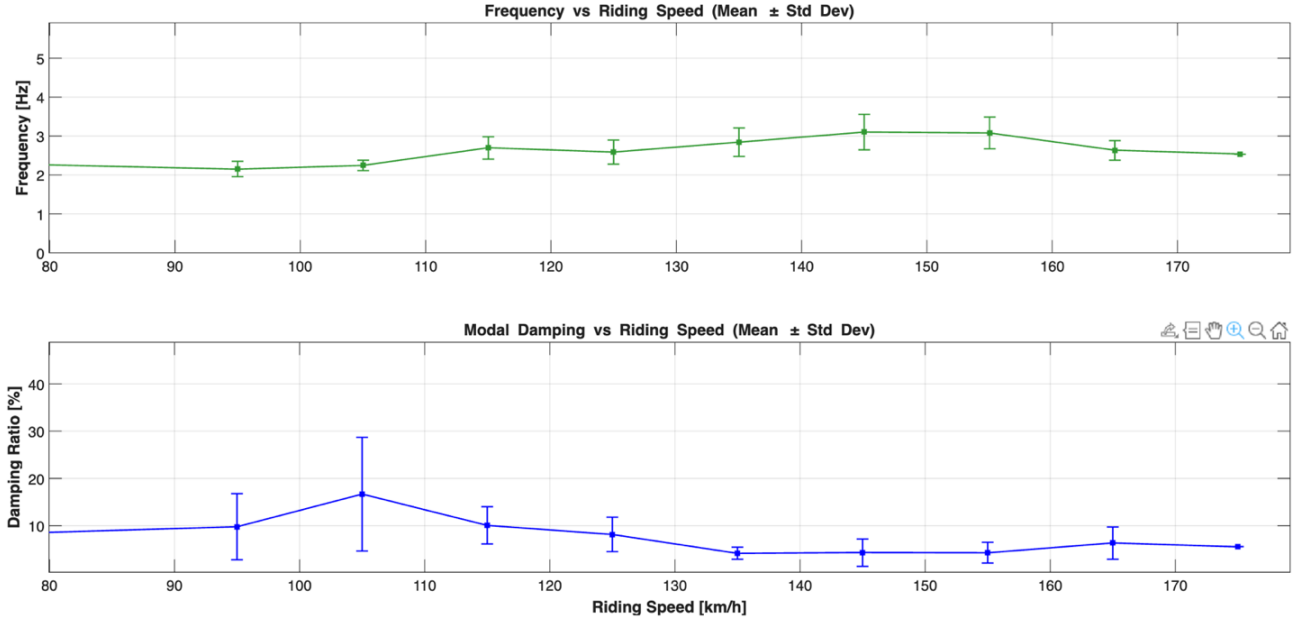


Figure 14: Clustered statistical analysis results

The tendency expressed by the plots respects the expectations, as a matter of fact the trend of the frequencies is floating around the value of 2.5 Hz, as we expect for the weave vibration, and the damping decreases while speed increase.

The standard deviation of the two trends shows the variability of the acquisition with respect to the noises and variations that could occur during the computation. Especially in the damping graph, this statistic interval decreases at high speeds, and this can be explained by noticing that at high speed the weaving becomes clearer because enters the critical zones and the mechanical parameters, that could ruin the measure, have less impact.

3.2.3 Gauss Diagram

The calculation of eigenvalues (λ) is the fundamental objective of experimental modal analysis, as these parameters uniquely characterize the system's dynamic stability independent of external inputs.

Physically, each eigenmode (such as the weave) is described by a pair of complex conjugate eigenvalues defined as:

$$\lambda = -\sigma \pm j\omega_d$$

In this composition, the real part (σ) corresponds to the decay rate, determining the speed of energy dissipation and the stability margin; the imaginary part (ω) represents the damped natural frequency, defining the periodicity of the oscillation.

By mapping these eigenvalues on the complex plane (Root Locus), the evolution of the motorcycle's stability across the speed range can be assessed instantaneously.

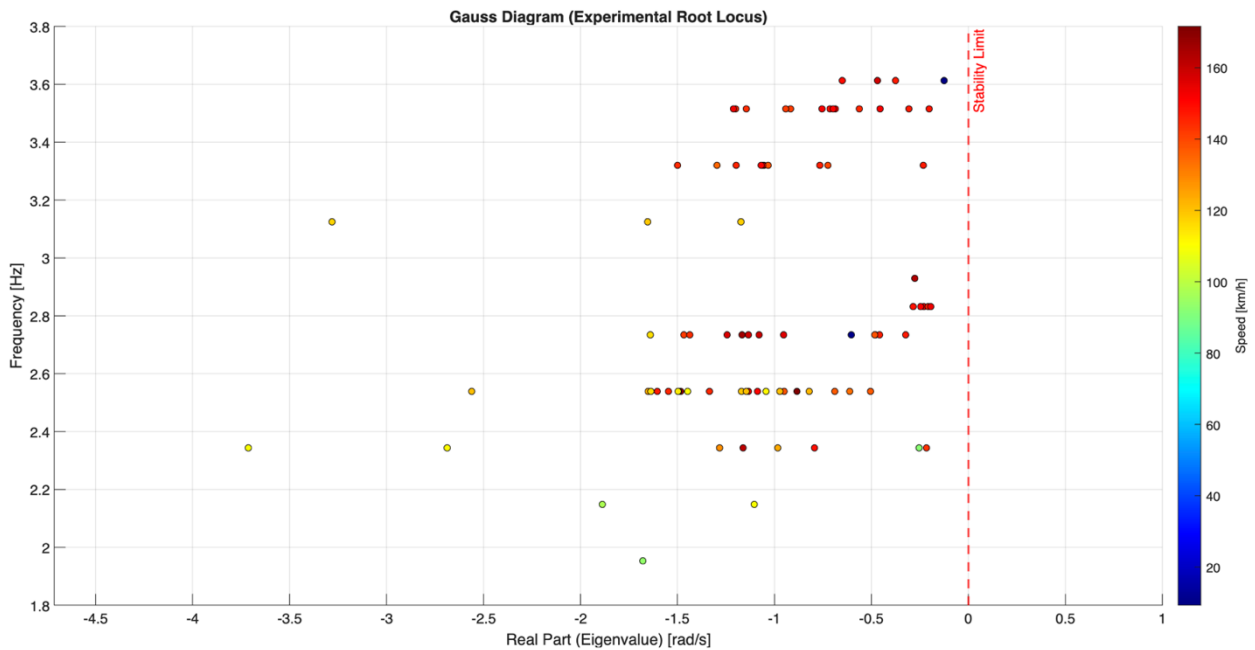


Figure 15: Experimental Gauss Diagram

In this plot, it is evident that the increase in speed is directly linked to the approach of the real part of the eigenvalues toward the critical threshold of 0 rad/s.

As a matter of fact, if the real part is $\sigma > 0$, the motorcycle behavior becomes unstable and, consequently, very dangerous for the rider, since the oscillations are no longer damped and diverges.

To provide a complete analysis, the plot in Figure 16 has been computed. It shows the imaginary part of the eigenvalues versus the real part and confirms the relationship between decreasing stability and increasing speed. The dashed lines represent damping percentages, illustrating that a reduction in damping corresponds to an increase in system instability.

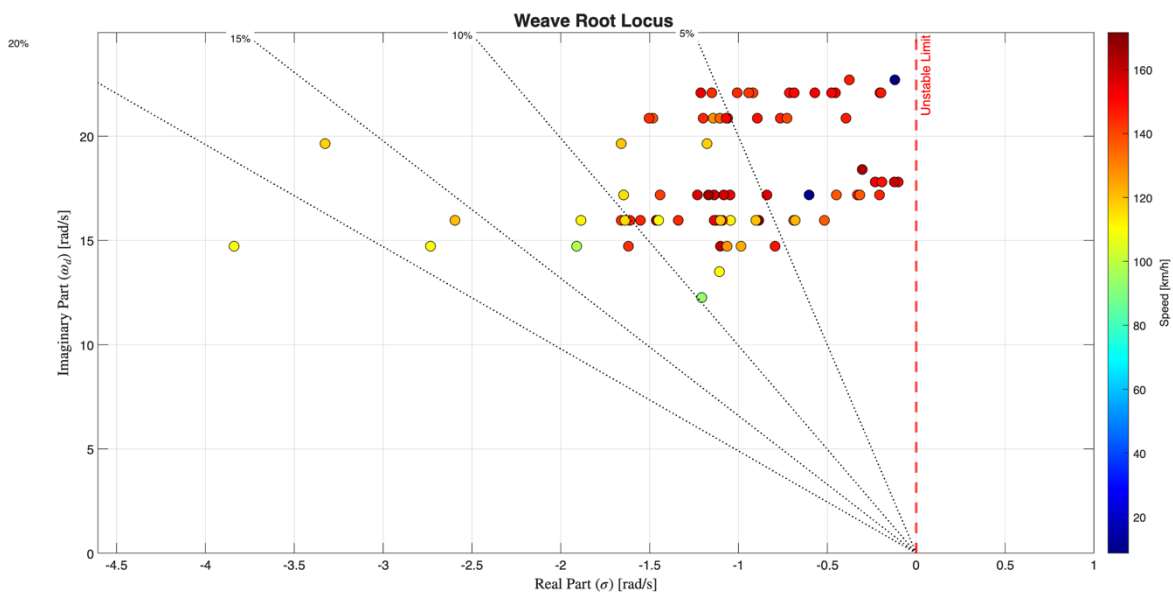


Figure 16: Root Locus - Damping ratio

At last, another graphical representation has been performed: in this case both axes are expressed in Hz and the alpha values (Real Part) are more compact, even if it expresses the same physical meaning.

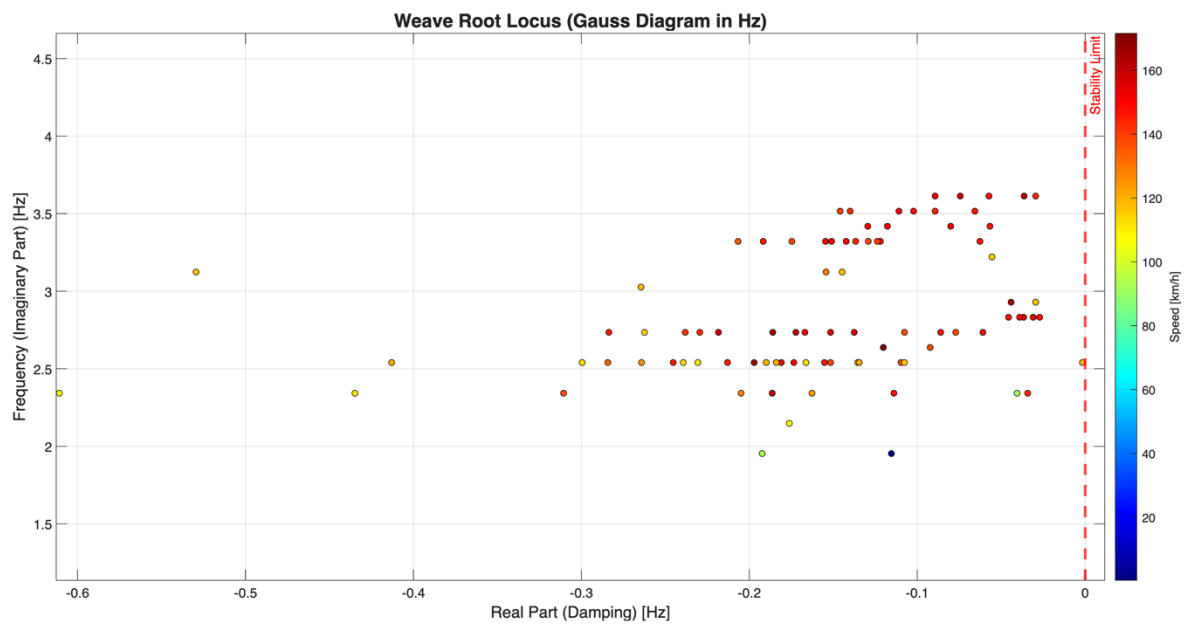


Figure 17: Gauss Diagram in Hz

3.2.4 Additional analysis: section cluster

In order to better characterize the differences between the sections, an additional graphical representation has been produced.

A clustered analysis of the trajectories was performed for the three most significant plots: speed-frequency, speed- damping ratio and Root Locus.

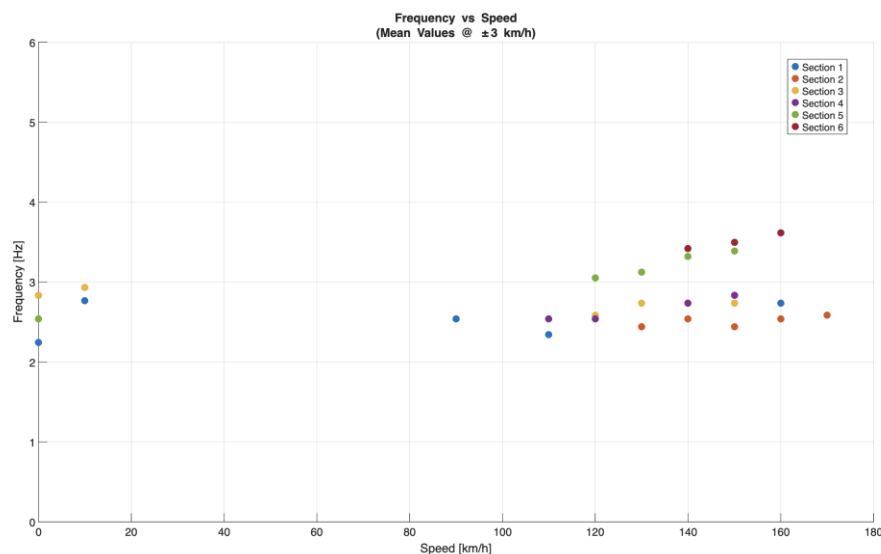


Figure 18: Clustered Speed-Frequency

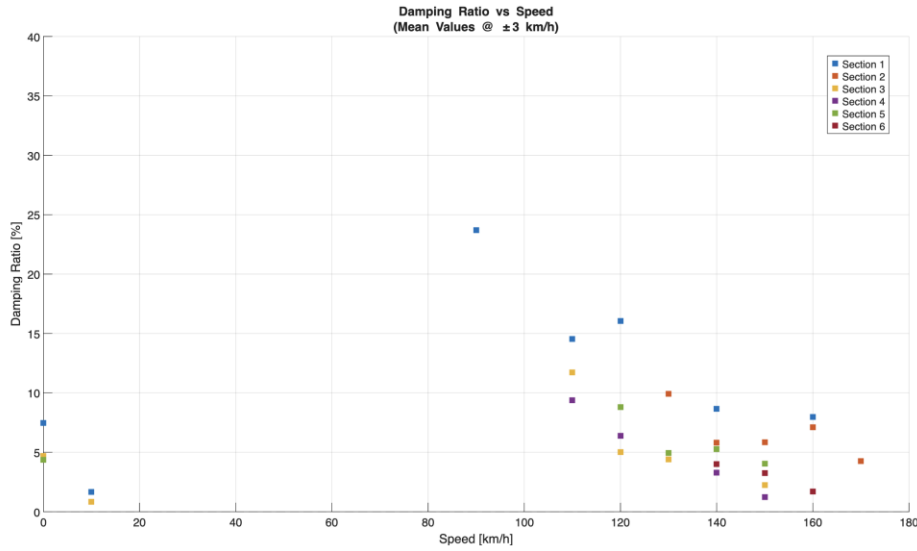


Figure 19: Clustered Speed- Damping Ratio

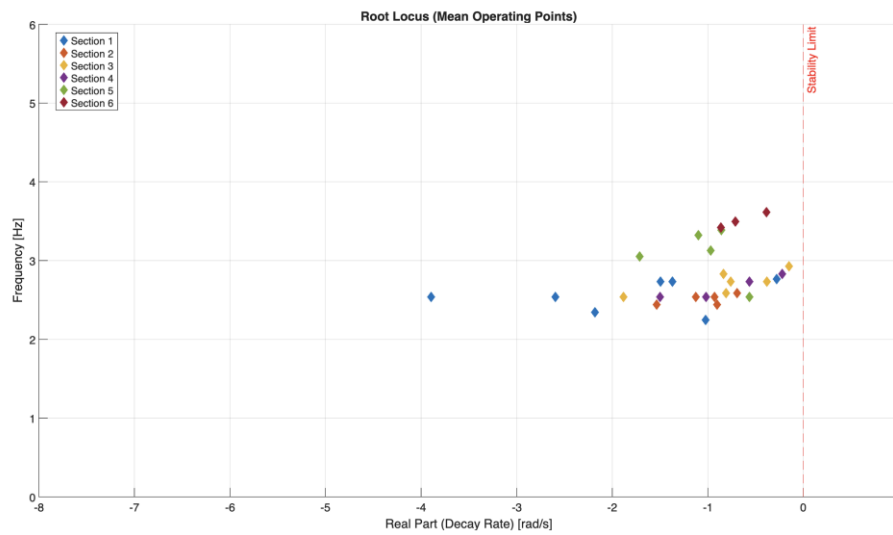


Figure 20: Clustered Root Locus

The graphs have been obtained by calculating the mean value of the parameter (frequency or damping ratio) detected in each section with a range of 3 km/h around the periodic values (10 km/h, 20 km/h...). This procedure aims to include a largest number of peaks to mean to guarantee a better final representation.

These plots confirm the previous observations about the differences between the beginning and the end of the experimental test.

In fact, section 5 and 6 highlights more drastic behaviors in terms of higher frequencies and lower damping ration, suggesting a setup modification.

3.2.5 Additional analysis: Wobble identification

A supplementary analysis of the wobble mode was conducted to detect latent oscillatory content and enhance the overall assessment of the experimental dataset. The steering angle signal underwent band-pass filtering within the specific 8–11 Hz frequency range, followed by a moving RMS (Root Mean Square) envelope calculation to quantify the phenomenon's intensity as a function of forward velocity.

The resulting stability map in Figure 21 confirms the presence of wobble peaks in the expected 60–80 km/h range; however, significant spectral content detected at higher velocities is attributed to acquisition artifacts or sensor noise rather than physical vehicle instability, as evidenced by the lack of correlation with the frequency-speed scatter plot.

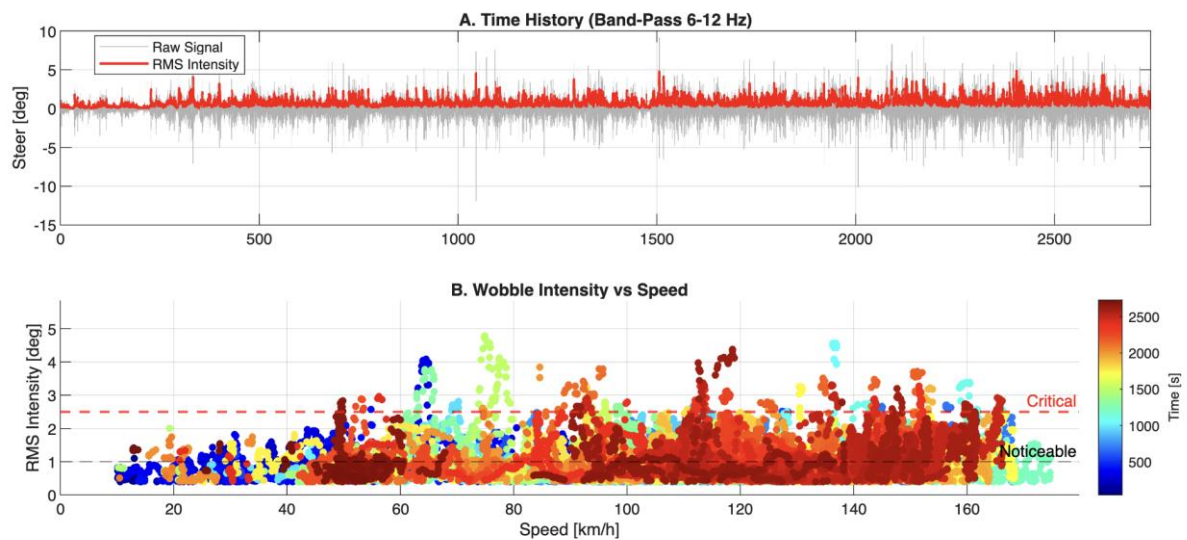


Figure 21: Wobble analysis

3.3 Conclusions

The analysis performed shows important results that are coherent with the expectations based on the state of the art review: weave is a oscillation characterized by frequencies between 2-4.5 Hz that can become critical at high speed, especially under low-load conditions.

In fact, the considerations over the plot results lead to the conclusion that the configuration of the bike has been probably changed during the acquisition (during section 3 and 5 there are stops), in order to highlight the different behaviours of the motorcycle. By the subsequent simulations the identification of a possible change in the setup will be executed.

Despite these changes, the graphs show that the stability of the bike is maintained during the path, thanks to the ability of the driver while managing the oscillations, together with the bike's damping instrumentation.

4. Numerical analysis

A 4-DOF numerical model has been used to evaluate a motorcycle behaviour in different configurations. These evaluations regard the motorcycle utilized by R.S. Sharp in 1971 for the analysis performed for the article [4].

Following a comprehensive analysis of the state of the art, the following key elements emerged as the main drivers of motorcycle stability in critical situations, such as response to vibrations, cornering performance, and response to road excitation:

1. Geometric and Mass Parameters:

- Center of Mass (CoM) Position: moving the rear frame CoM forward or lowering it improves Weave damping.
- Wheel Inertia: increasing rear wheel inertia enhances high-speed Weave stability.
- Steering Inertia: a critical factor for Wobble; increasing it reduces vibration frequency but can affect high-frequency stability.
- Steering Geometry (Rake): a steeper steering head angle improves low-speed stability but reduces Weave damping at high speeds.

2. Structural Properties:

- Front Fork Stiffness: the primary structural factor for Wobble; fork flexibility is destabilizing at low speeds.
- Frame Stiffness: torsional stiffness is crucial; insufficient stiffness compromises high-speed Weave stability.

3. Tires and Damping:

- Steering Damper: the most effective component for stabilizing the Wobble.
- Tire Relaxation Length: the delay in lateral force generation; larger relaxation lengths tend to destabilize both Weave and Wobble.

4. Rider Influence:

- Rider Body Damping: the rider's body acts as a passive damper, significantly stabilizing the Wobble mode.
- Load (Suspended Mass): varying the load mainly affects the Weave mode, while having a negligible effect on Wobble frequencies.

As the model allows different kinds of simulations and comparisons, different kinds of setups will be analyzed.

4.1 Sensitivity test

4.1.1 Geometrical parameters

To promote a complete weave analysis, three geometrical parameters have been analysed: steering damper, front mass centre, wheel base.

The first element considered is the **steering damper**: a device (usually hydraulic) that dissipates energy associated with steering rotation. It connects the frame to the steering system (forks/crown clamps) and resists the rotational speed of the handlebars. By varying its impact value of $\pm 20\%$ we can observe some changes.

The lowering of the steering damper is linked to a slightly higher stability, while the increase of the damper provokes the opposite effect. The reason depends on the fact that the reduction of this parameter delays the corrective steering response required for weave stabilization, alters critical phase relationships among steering, yaw, and roll motions, and reduces the effectiveness of front-tire-generated stabilizing forces. As a result, the modal damping of the weave mode decreases, potentially leading to instability at certain speeds.

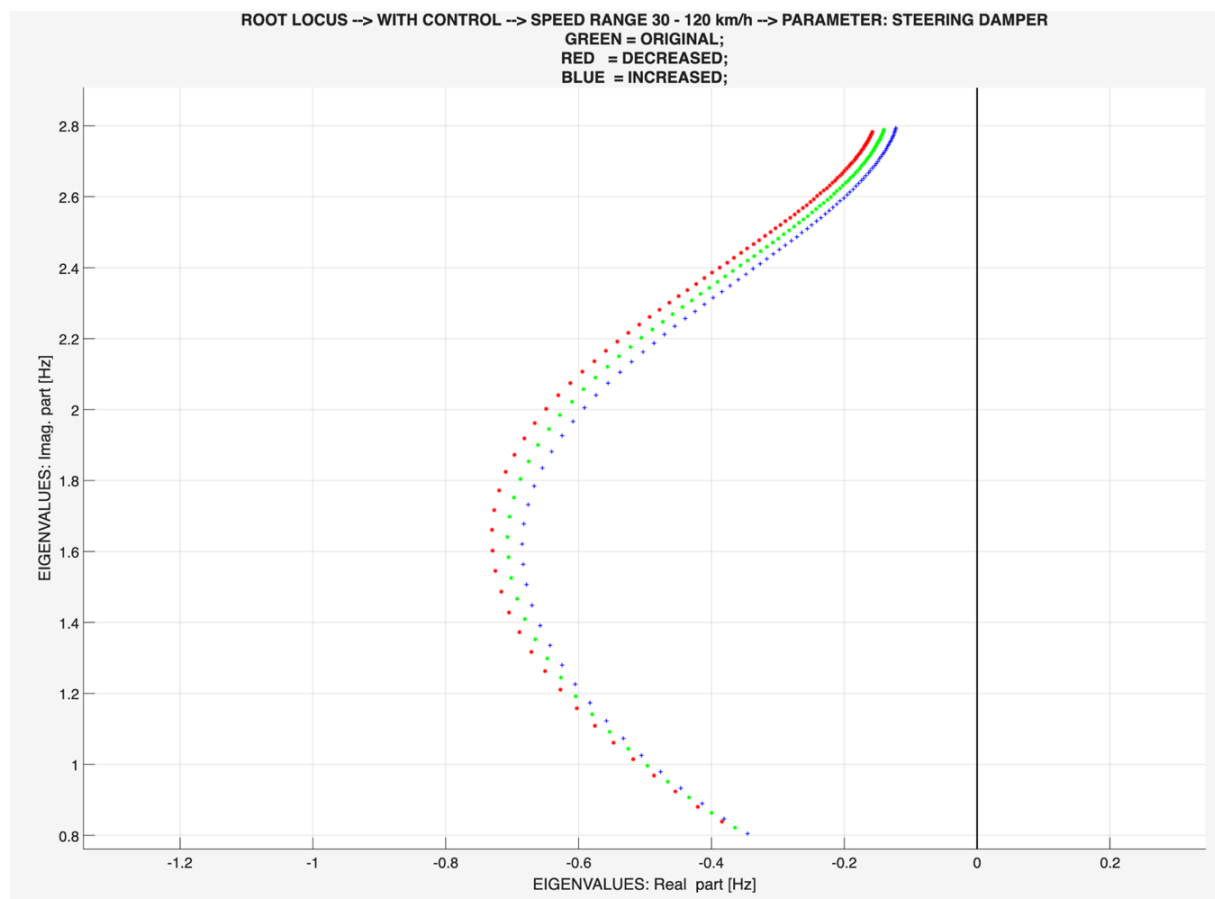


Figure 22: Gauss Diagram: Steering Damper $\pm 20\%$

However, the plot does not highlight important variations of stability linked to this parameter, so we can conclude that this is an almost negligible parameter.

The second parameter modified is **the front mass centre** that refers to the center of gravity of the entire steerable front assembly that includes the front wheel, fork, handlebars and associated fittings. The height has been modified by +/- 40%.

The Gauss Diagram shows that at high speeds (around 130 km/h) the bike's behavior is almost the same, it gets slightly better at low speeds (around 20 km/h) and improves the stability properties at medium speeds.

A lower center of mass optimizes the feedback loop between steering and lean angle by minimizing inertial delay. In contrast, a high center of mass introduces a phase lag that disrupts natural damping, significantly reducing stability during the mid-speed range.

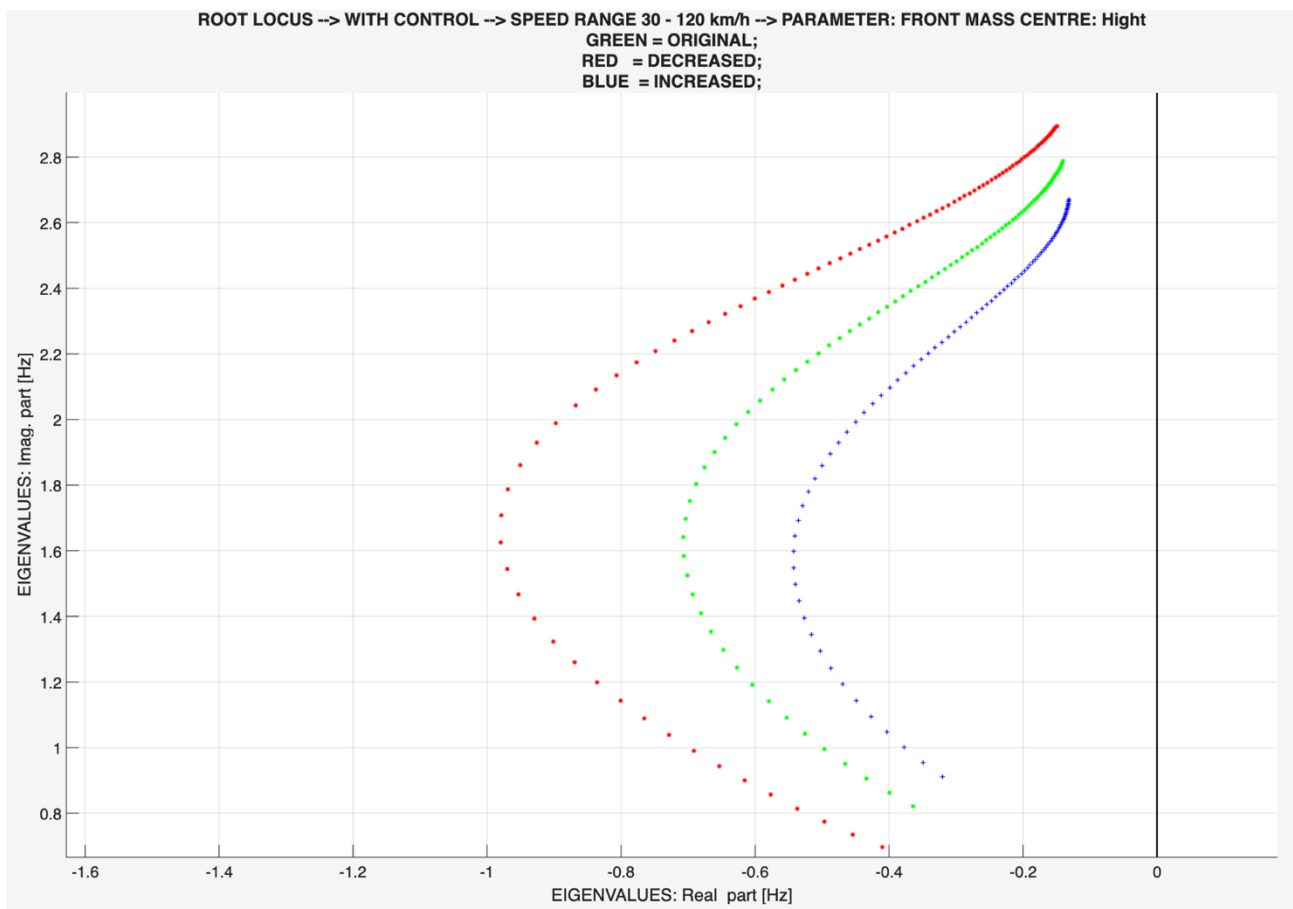


Figure 23: Gauss Diagram: Front Mass Centre +/- 40%

The last parameter analyzed is the **wheel base**: the distance between the two wheels. The value has been modified by +/- 40%.

This parameter strongly influences the stability of the bike, as demonstrated by the root locus diagram. At high speeds an increased wheel base leads to instability, while at low speeds the impact of this variation is almost absent. At mid-speeds the curves show a significant improvement in stability.

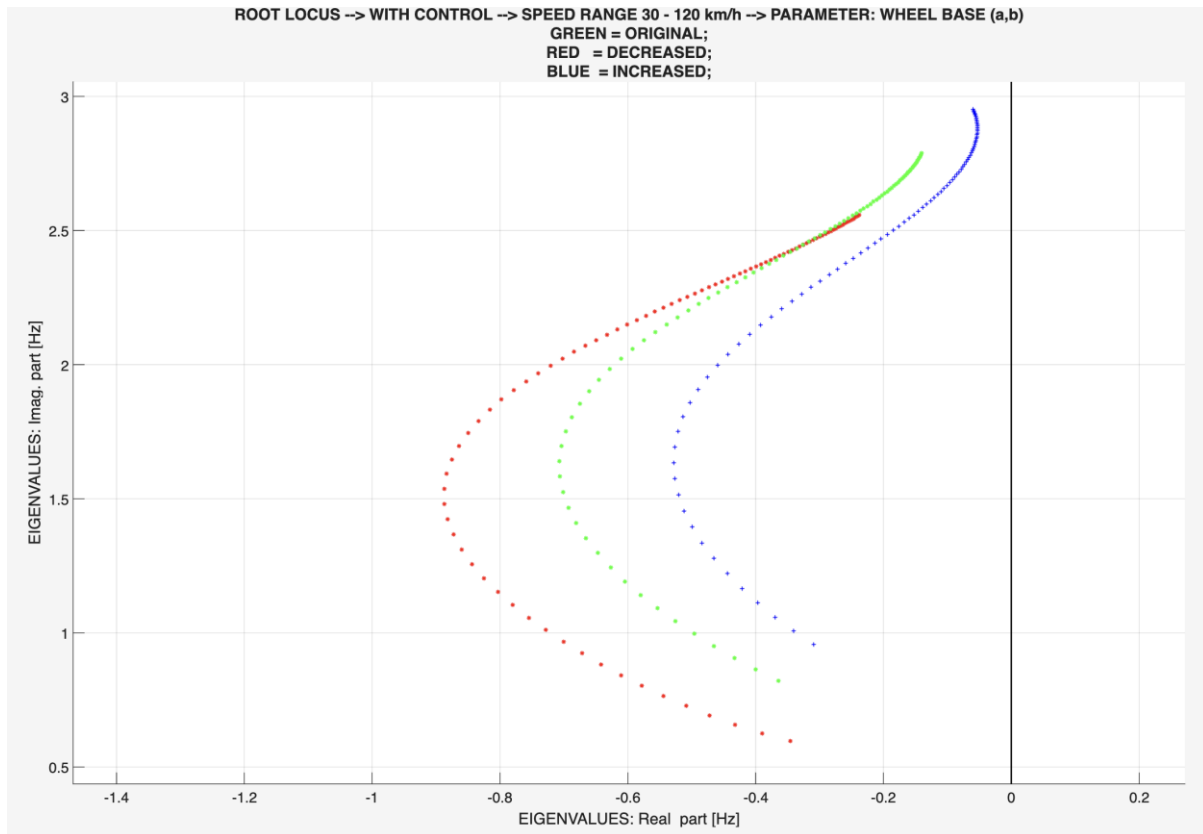


Figure 24: Gauss Diagram: wheel base +/- 40%

4.1.2 Load conditions

Together with geometric conditions, also load variations can modify bike's stability at different speeds. For this analysis the same 4-DOF model and the same motorcycle model (R.S Sharp model used in 1971) have been used.

The model offers different load configurations to test and the first one compares the bike in its unloaded condition with the bike carrying only the rider's weight; the result of the simulation is reported in the Figure 19 below.

The analysis reveals that both configurations exhibit reduced stability margins at low and high velocities, while achieving peak stability at intermediate speeds. Specifically, the rider (75 kg)-load configuration displays more polarized behavior across the speed range: at low speeds, the rider's mass slightly raises the vertical center of gravity (CoG), which increases the rolling moment but is easily managed by the rider's active balance, at higher velocities, the rider's body provides significant aerodynamic damping and increases grip with the ground, which helps dissipate energy from the weave mode.

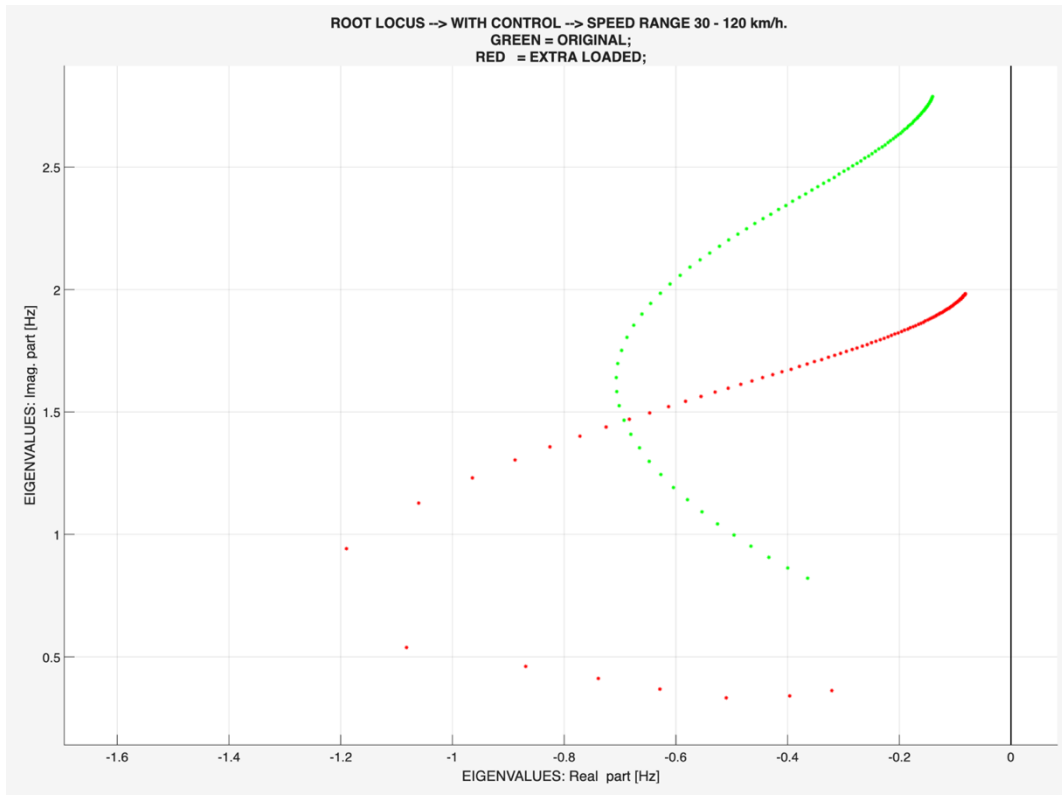


Figure 25: Gauss Diagram: Bike vs Bike+ rider (75kg)

The second simulation imposed a +10 kg load on the rear in addition to the driver's load. The results are reported in the plot below and confirm the previous description.

As a matter of fact, at higher speeds the behavior is slightly more unstable, while the stability increases significantly at mid-range speeds and the oscillation frequencies are a little lower at high speeds

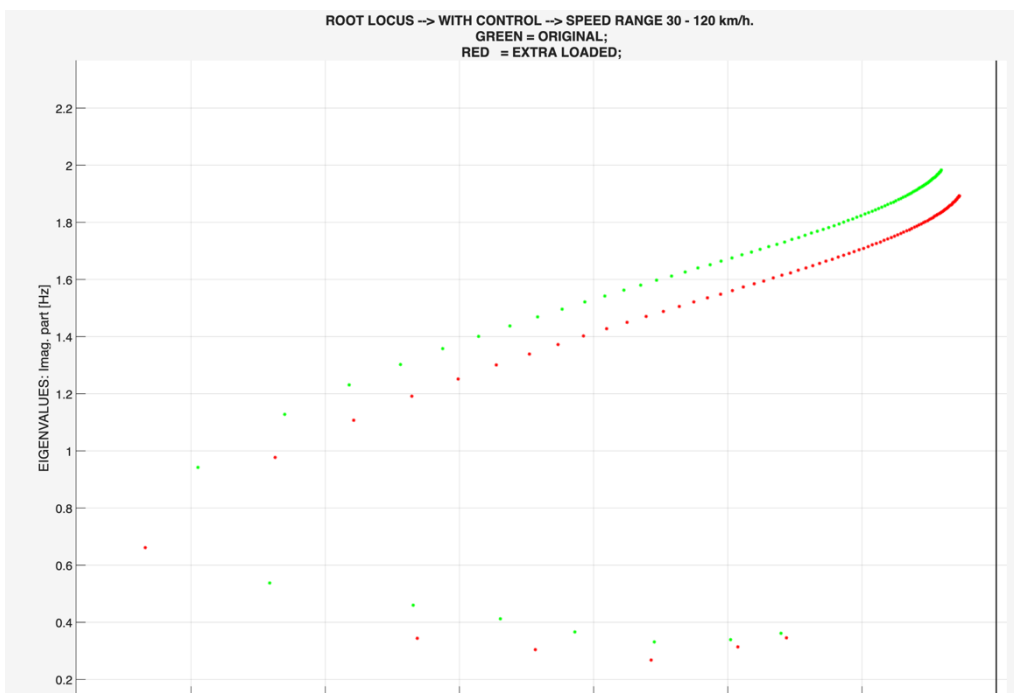


Figure 26: Gauss Diagram: Bike + rider vs Bike+ rider+ 10kg (rear)

To make a comparison, a simulation of +15 kg load addition on the front of the bike was performed.

The results are quite similar to the ones obtained with the previous analysis: in this case the stability slightly increases, especially at low speeds.

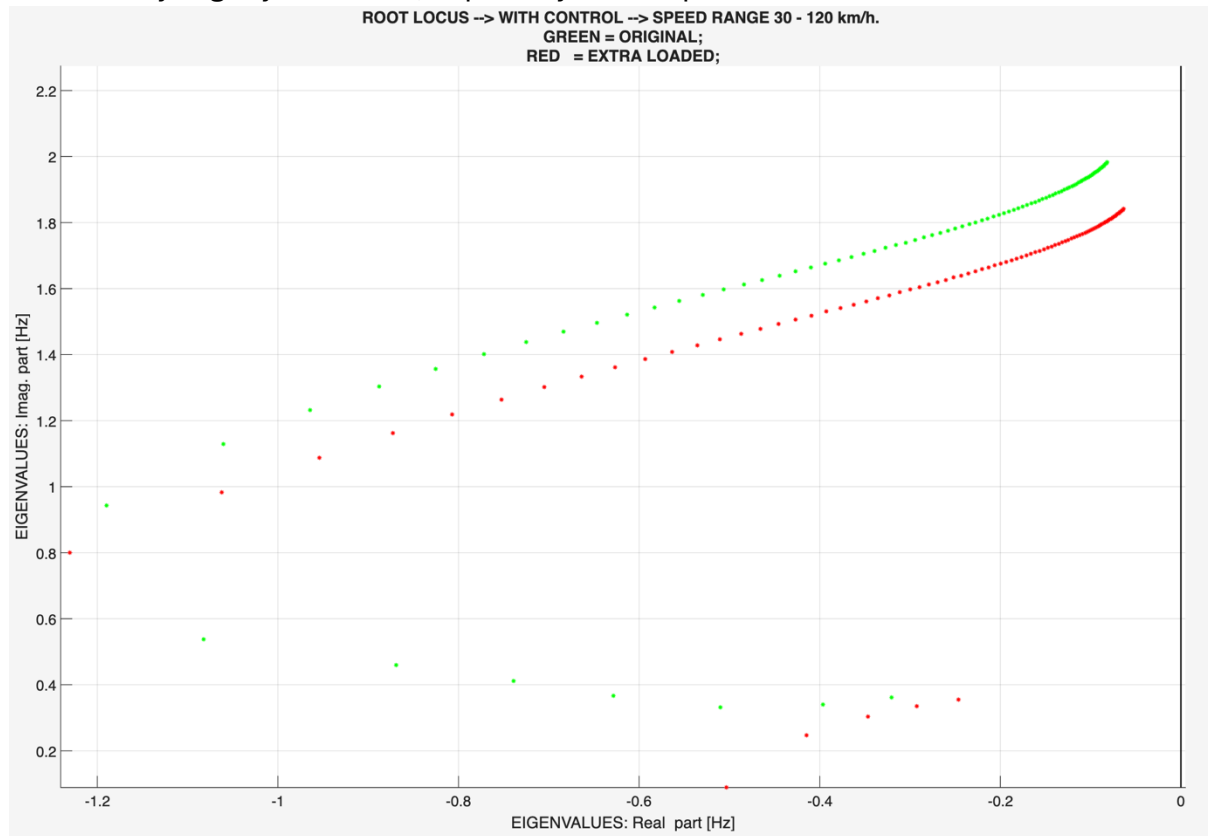


Figure 27: Gauss Diagram: Bike+rider vs Bike+ rider+ 15 Kg (front)

The last simulation includes many kinds of loads: driver + passengers + bags. This setup highlights a dangerous situation of instability at the speeds around 120 km/h, given by the extra-weight added.

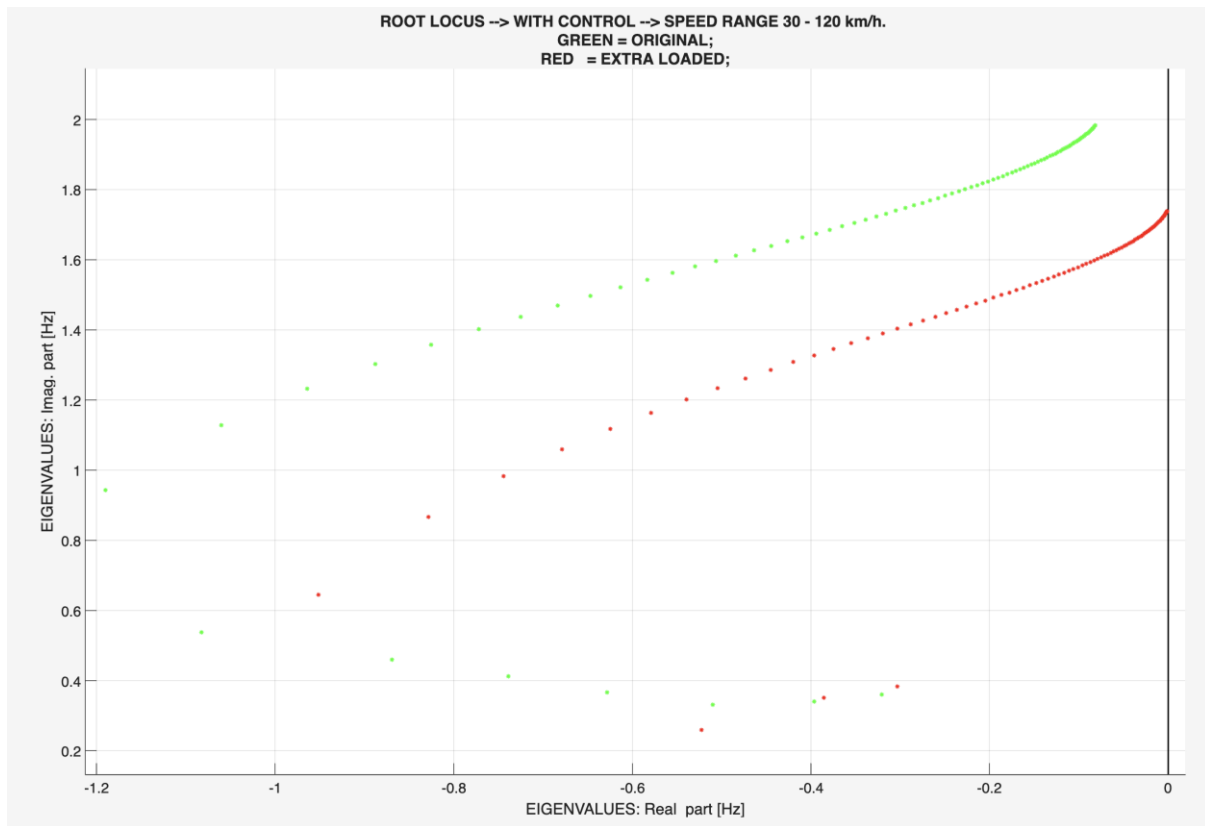


Figure 28: Gauss Diagram: Bike+rider vs Bike+rider+passenger+bags

4.2 Time domain analysis

After the evaluation of different parameters on the motorcycle's stability, the next step was to proceed with a time domain simulation in order to compare different kinds of setups.

To run the test the 4-DOF model has been used.

The first analysis refers to the basic setup adopted in the previous chapter, that establishes the standard parameters before any variation, and the speed was set to 120 km/h: the critical value characterized in the sensitivity tests.

The test was run considering an impact during the drive, the most similar event referring to the experimental analysis and real-life experience.

The driver, as soon as the impact occurs, tries to minimize the effect of the oscillation by using his own body as damping method.

4.2.1 Standard setup

The simulation of the basic setup led to the subsequent results related to the sensors acquisitions:

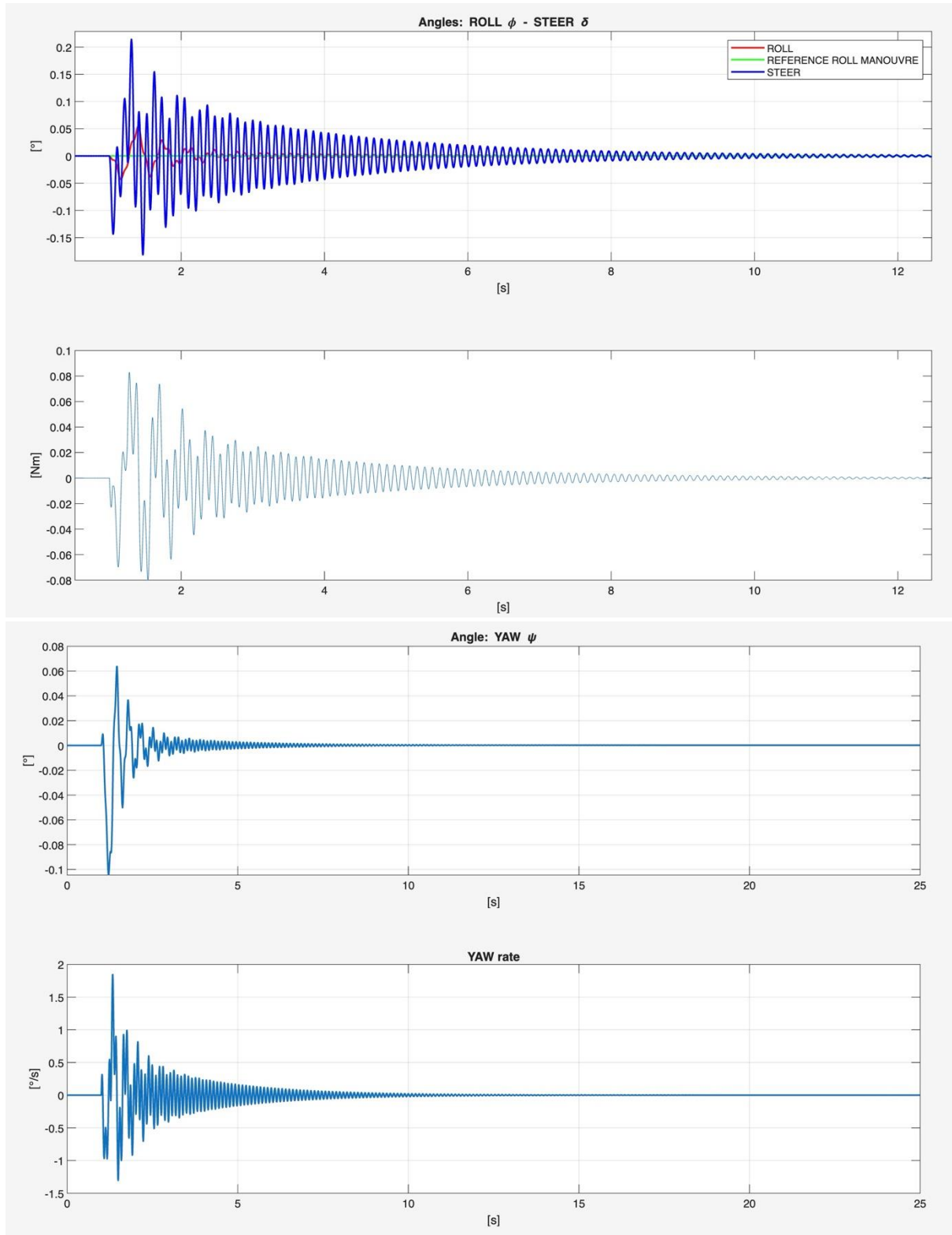


Figure 29: Sensors acquisitions – standard setup

The curves show that the response to the impact is an exponential damping, as we expect, that lasts for some seconds before can be considered negligible.

A representation of the lateral force contact was also supplied, in order to observe the differences between rear and front behaviour:

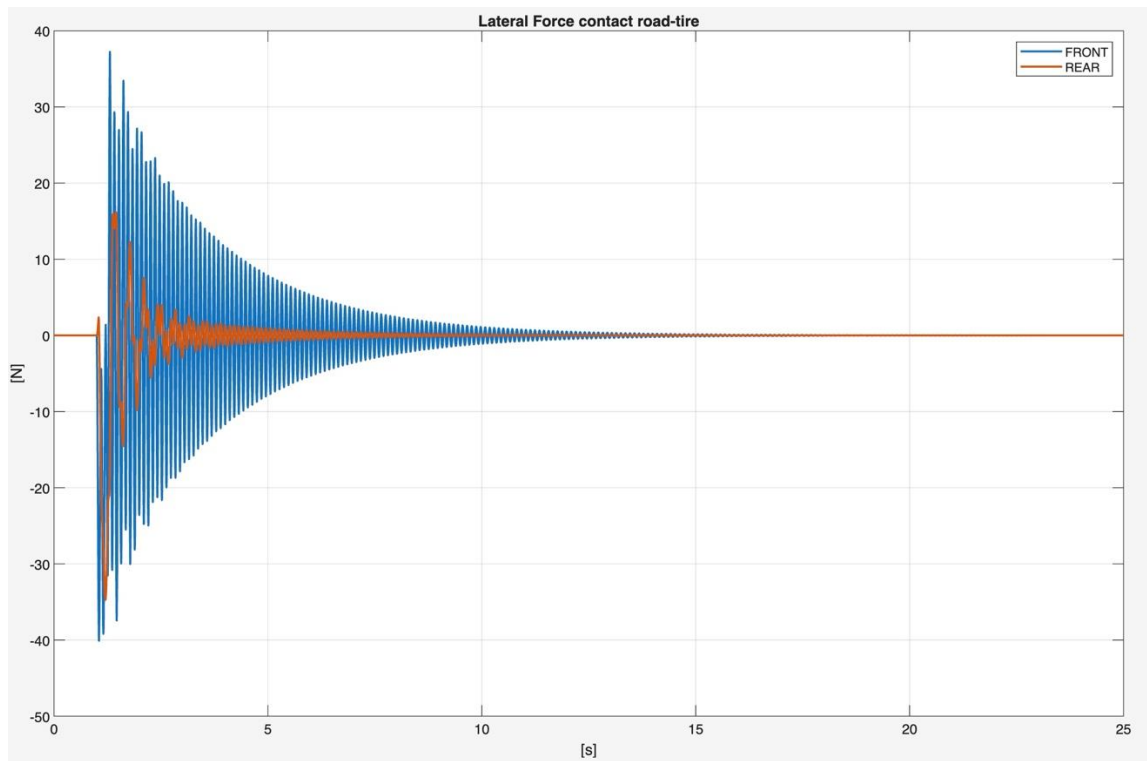


Figure 30: Lateral force contact (Front vs Rear) – standard setup

Lastly, the most important representation is the spectral diagram of the impact response, that exhibits the frequency-domain characterization of the vehicle's dynamics.

In particular, the two main peaks that appear on the plot are related to the weave mode ([2,4] Hz) and the wobble mode (9.5 Hz). During this analysis, as in the previous ones, the focus will be on the weave mode.

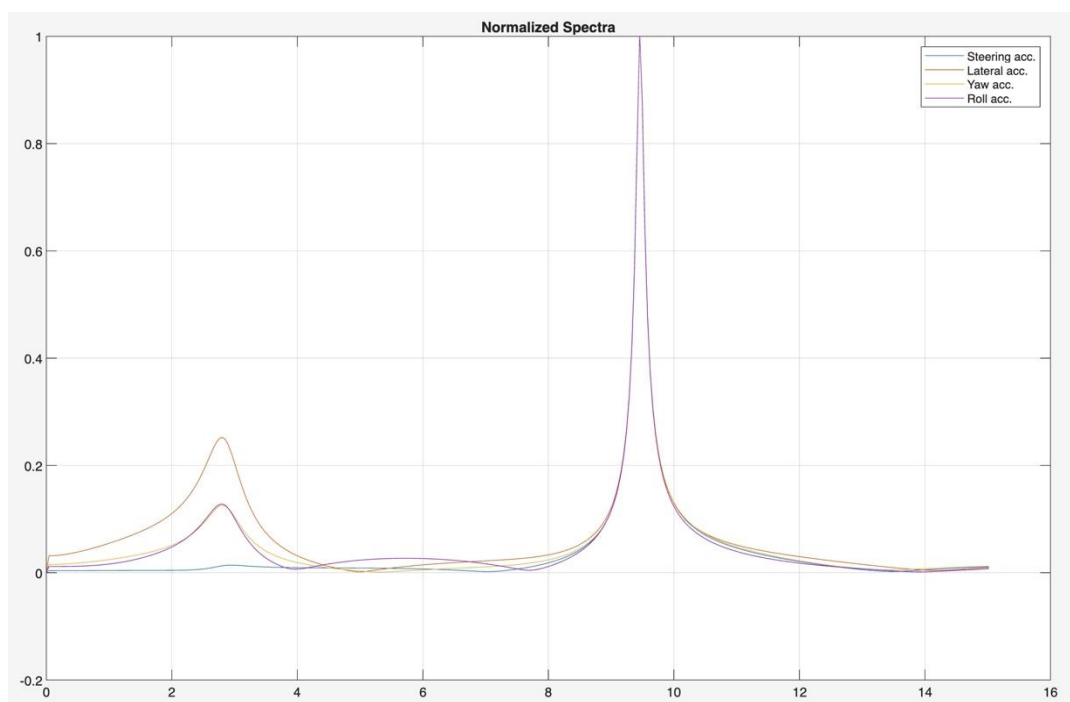


Figure 31: Normalized spectra showing weave and wobble modes – standard setup

4.2.2 Modified parameters

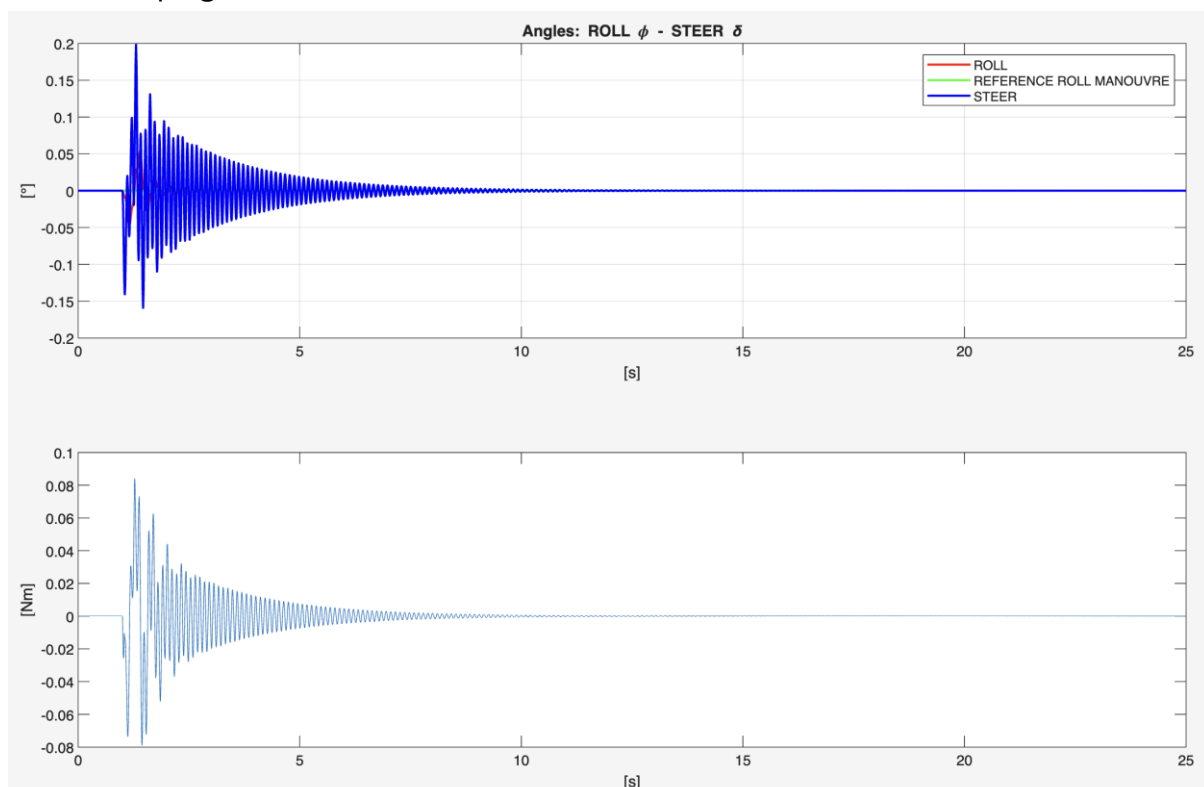
In this section some parameters will be modified to try to stabilize the weave mode.

The parameters taken into account are:

1. a : semi-wheel base: rear centre of mass and front tire contact.
2. tt : front trail
3. h : height of rear centre of mass
4. hf : height of front centre of mass
5. I : inertia momentum
6. C_{δ} : steering damper

Try	Geometry	Mass	Inertia	Damping (N*s/m)
1	a -10%	Front mass +10% Rear mass -5%	$I_{rear,z}$ +5% $I_{front,z}$ -5%	$C_{\delta} = 3.5$
2	tt +3% h -10% hf -10% a -10%	Front mass +10% Rear mass -5%	/	/

Try 1 led to the subsequent results that clearly show an improvement of the weave mode damping.



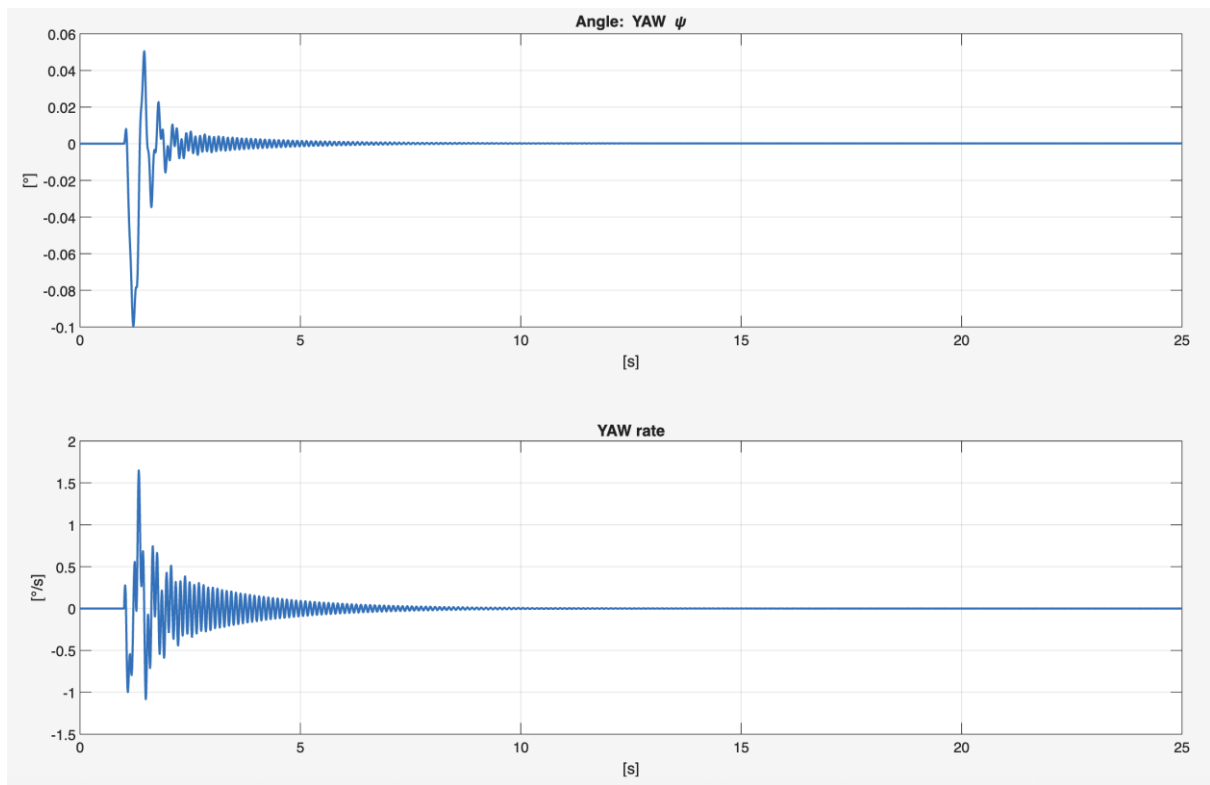


Figure 32: Sensors acquisitions -Try 1

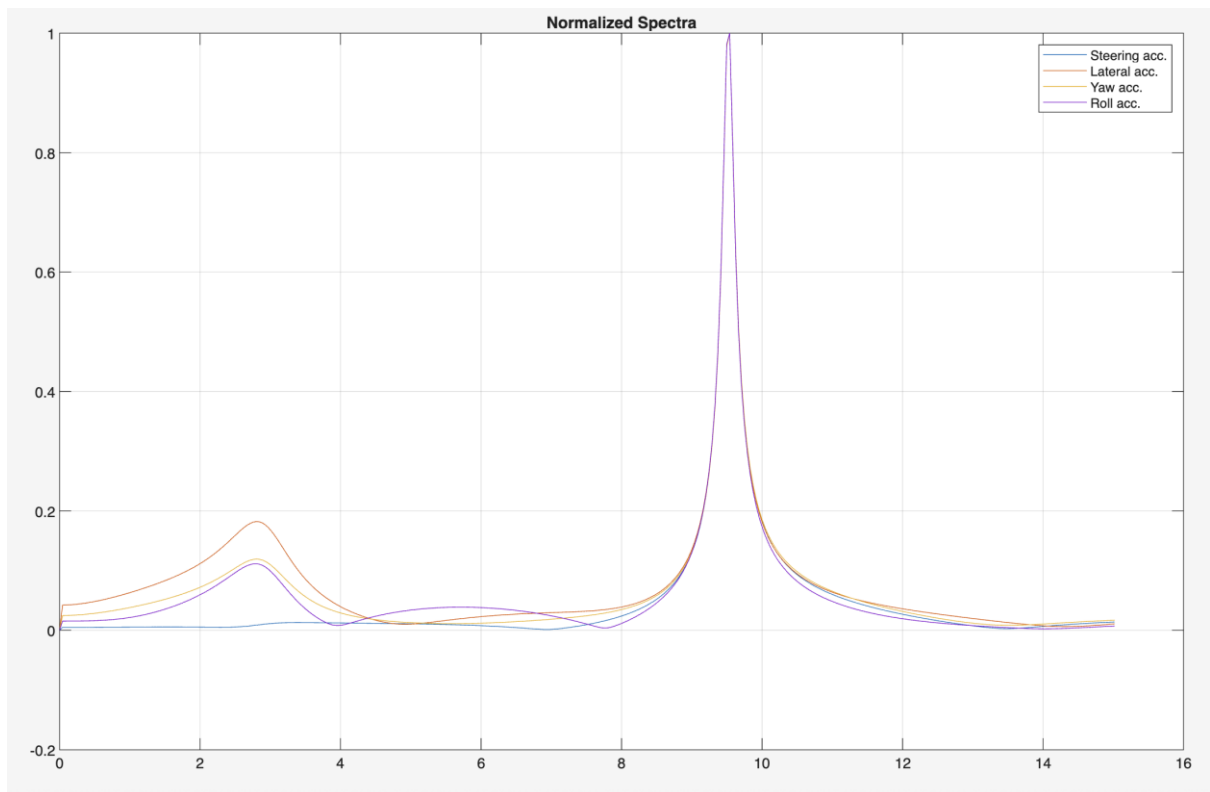


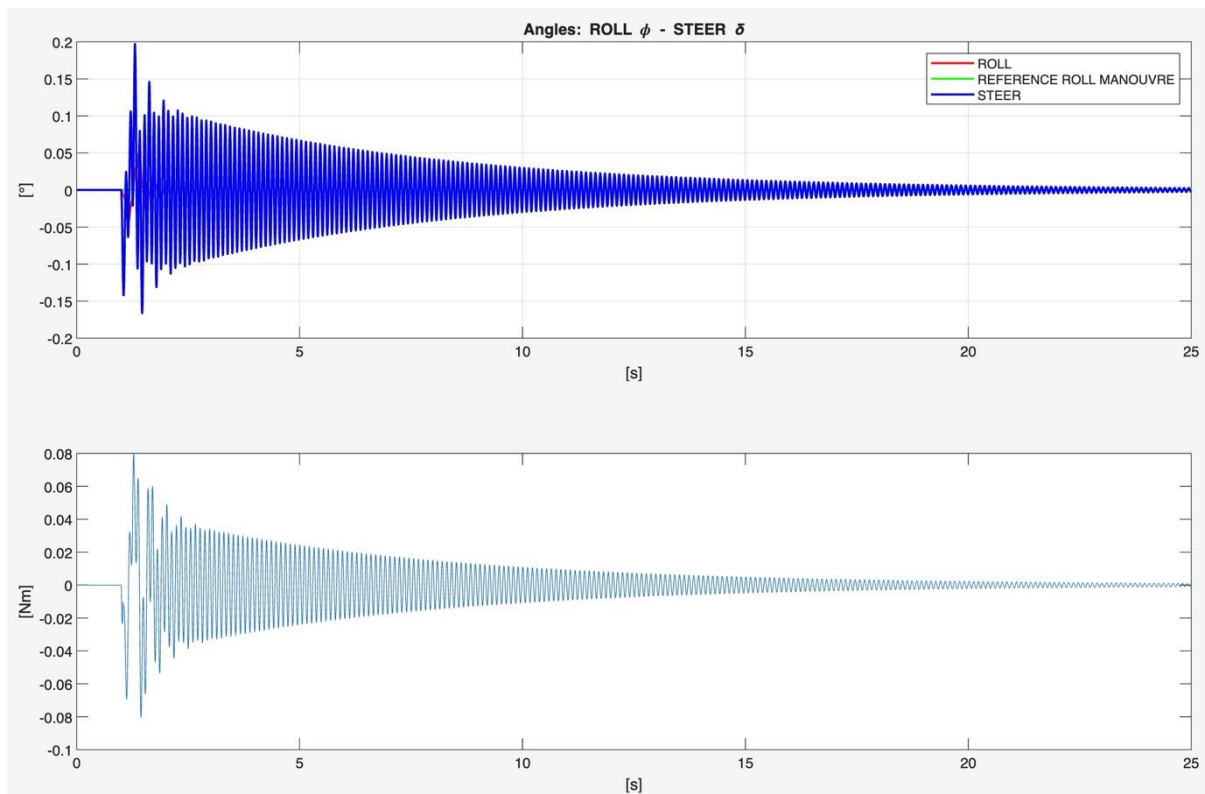
Figure 33: Normalized spectra - Try 1

Other setups have been tried in order to reduce in a drastic way the weave peaks, but the results exhibited divergent impulse responses, even if the normalized spectra were completely flat in the interested bandwidth.

The final decision was to keep Try 2 as the best compromise between all the parameter modifications.

This setup, in fact, reduces significantly the weave mode vibrations and shows a convergent and stable sensors response, despite the higher dissipation time.

In particular, modifying moments of inertia disrupts the delicate balance between the vehicle's kinetic energy and its damping capacity. Physically, excessive inertia introduces a phase lag between the perturbation and the corrective tire forces, preventing the system from synchronizing its self-aligning torques. This delay transforms stabilizing feedback into positive reinforcement, causing the weave and wobble modes to extract energy from the forward motion and leading to divergent, unstable oscillations.



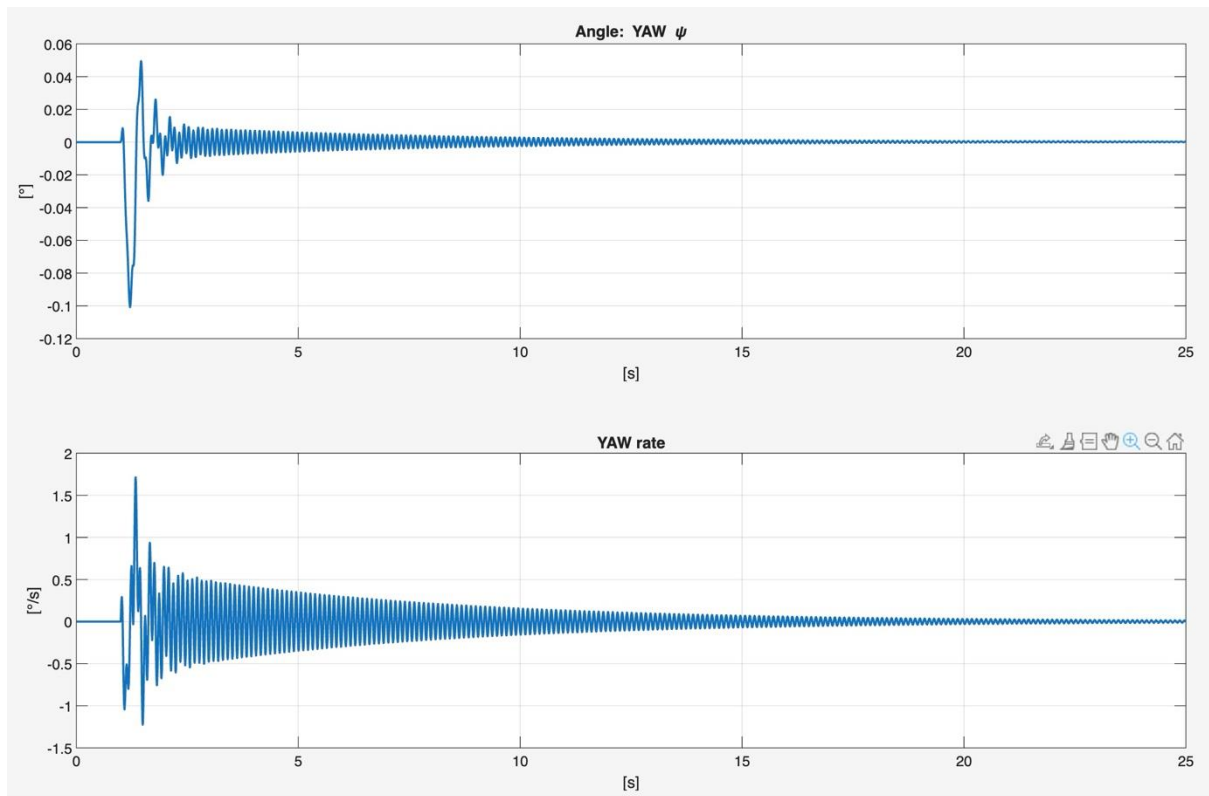


Figure 34: Sensors acquisitions - Try 2

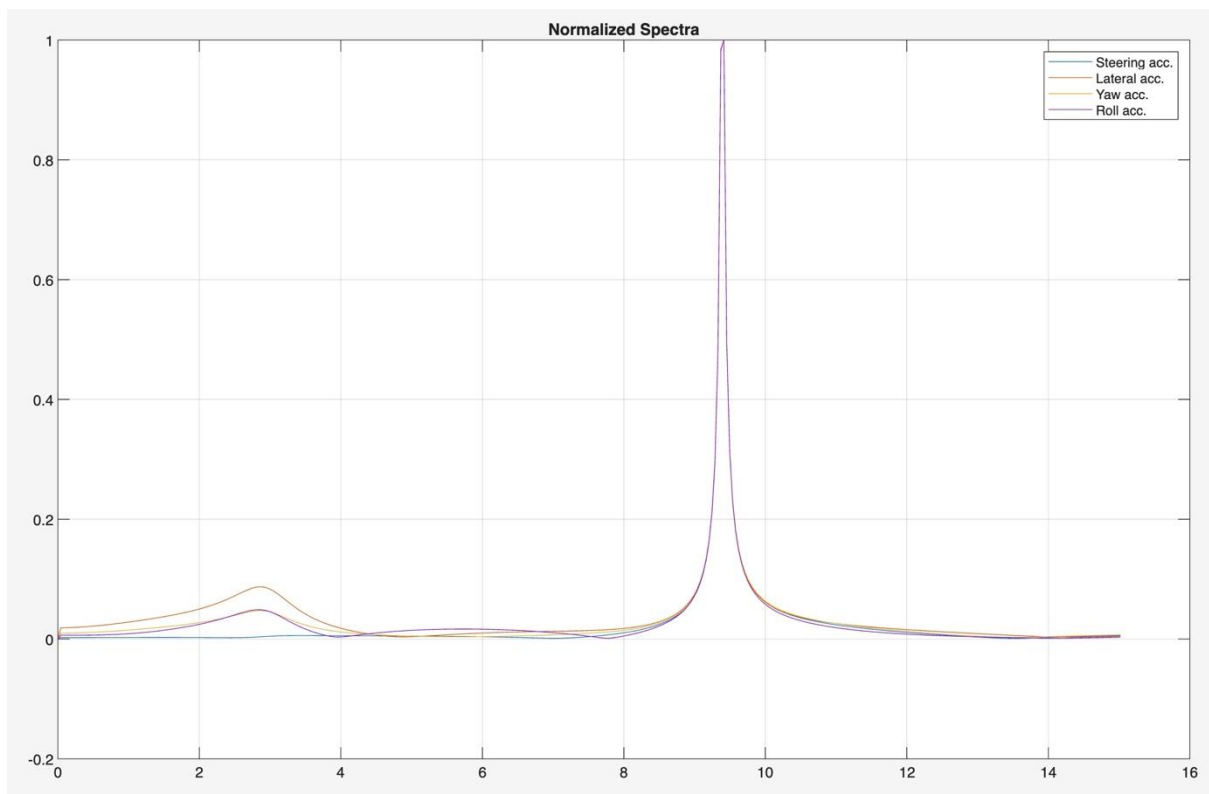


Figure 35: Normalised spectra - Try 2

4.3 Conclusion

The overall conclusion that can be obtained is that this study has successfully validated the dynamic behavior of a motorcycle that reacts to impulsive excitations, providing a deeper understanding of the factors governing the motorcycle's stability.

The experimental analysis was based on real-road data acquired using MEMS gyroscopes and GPS sensors. Controlled rider-induced excitations allowed the identification of transient responses associated with the weave mode. Advanced signal processing techniques, including filtering, Hilbert transform reconstruction, Fourier analysis, and statistical aggregation, were employed to estimate damped natural frequencies, decay rates, and modal damping ratios. The results showed frequencies consistently within the expected 2–4.5 Hz range and a clear reduction in damping with increasing speed. The Gauss diagrams provided an effective visualization of the progressive loss of stability, as eigenvalues moved toward the imaginary axis at higher velocities. These findings are fully consistent with the state of the art and confirm the reliability of the experimental methodology.

After the computation of all the diagrams it is evident that some changes have been adopted during the execution of the acquisition, that could explain the shifts that appear on the graphs related to the time gradient.

In particular, the most likely option regards the load-reduction of the bike during the last sections of the path, especially during the two brief stops performed around 1500s and 2000 s.

The adoption of the 4-DOF numerical model made it possible to verify that the observed variations in damping and frequency trends are consistent with changes in both load conditions and geometric parameters. Parametric simulations performed starting from a pre-set and validated motorcycle configuration showed that modifications in mass distribution and geometry can produce stability variations comparable to those measured experimentally.

The sensitivity analysis on geometrical parameters and load conditions highlighted the critical trade-offs in motorcycle setup, while time-domain simulations revealed that while several configurations could theoretically flatten the normalized spectra, they can also lead to divergent impulse responses.

In conclusion, this work demonstrates the effectiveness of combining experimental modal analysis with numerical simulations to study motorcycle stability. The results underline the critical importance of the weave mode in defining safe operating conditions and emphasize how design choices, vehicle setup, and loading significantly affect dynamic behavior.

5. Bibliography

- [1] Sharp, R. S. (2001). The stability and control of motorcycles. Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science, 215(10), 1095-1107.
- [2] Passigato, F. (2015). Study of motorcycle dynamics for the improvement of the stability of weave and wobble eigenmodes [Doctoral Thesis, University of Padua].
- [3] Dębowski, A. (2017). Analysis of the effect of mass parameters on motorcycle vibration and stability. Archives of Control Sciences, 27(2), 231-247.
- [4] Sharp, R. S. (1971). Stability, control and steering responses of motorcycles. Journal of Mechanical Engineering Science, 13(5), 316-330.